CRICKSTATS

Cricket Performance Analytics System

BACHELOR OF COMPUTER SCIENCE

Submitted by

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Abstract

Abstract

CrickStats is a Salesforce-based cricket performance analytics application developed to manage, store, and analyse player statistics across multiple matches. The system is built using the Salesforce Lightning Experience and utilizes declarative features such as custom objects, master-detail relationships, roll-up summary fields, formula fields, validation rules, reports, dashboards, and flow automation.

The application automatically calculates important performance indicators such as batting strike rate, bowling economy rate, player averages, and overall performance scores. Validation rules help maintain accurate and reliable data, while the custom Lightning App interface improves usability and navigation for users. This project highlights the practical use of Salesforce data modelling, automation, and reporting tools to create an efficient cricket performance management system for real-world scenarios.

Introduction

Introduction

Cricket analytics plays an important role in assessing player performance and supporting teams in making strategic decisions. Monitoring statistics such as runs scored, wickets taken, strike rate, economy rate, and match averages helps coaches and analysts understand player strengths and identify areas that require improvement.

Traditional methods of maintaining these records manually are often slow, inefficient, and prone to human errors. As the need for data-driven decision making continues to grow, there is a requirement for an automated system that can store structured player information and generate analytical insights in real time.

The **CrickStats** system was developed to meet this requirement by using Salesforce declarative tools to create a centralized platform for managing cricket performance data. The system automatically computes key performance metrics, maintains data accuracy, and displays useful insights through interactive reports and dashboards.

The main objectives of the system are:

- To store and manage player performance information
- To automate the calculation of important cricket statistics
- To maintain data accuracy using validation rules
- To present performance insights through reports and dashboards
- To provide an interactive user interface using flow automation

Problem Statement and Description

Problem Statement and Description

Evaluating cricket player performance requires accurate statistical data from multiple matches. When these records are maintained manually, the process becomes slow and increases the chances of human errors. Coaches and analysts rely on correct performance data to assess players, but traditional record-keeping methods often create inconsistencies and make it difficult to compute advanced metrics such as strike rate, bowling economy rate, and overall performance rankings.

In addition, the absence of a centralized system makes it challenging to retrieve historical data and compare player performances effectively. Because of this limitation, an automated and well-organized system is necessary to store match details, perform statistical calculations automatically, and present useful insights through visual reports and dashboards.

The **CrickStats** application solves this problem by providing a Salesforce-based automated platform that efficiently manages cricket performance information and generates meaningful analytical results.

Feasibility Study

Feasibility Study

A feasibility study helps determine whether the proposed system can be successfully developed and implemented in a practical manner.

- **Technology Feasibility**

The project is technically viable because it is developed using Salesforce Developer Edition. This platform provides various built-in tools for data modelling, automation, reporting, and analytics. Salesforce also supports declarative development, allowing complex system logic to be implemented through configuration without extensive programming.

- **System Usability Feasibility**

The system is designed to be simple and easy for users to operate within the Salesforce Lightning interface. Users can easily create player records, enter match statistics, and analyse performance data without needing advanced technical knowledge.

- **Cost Feasibility**

The project is economically practical since Salesforce Developer Edition is available free for educational and development purposes. Therefore, the application can be implemented without additional infrastructure or software costs.

Team Creation

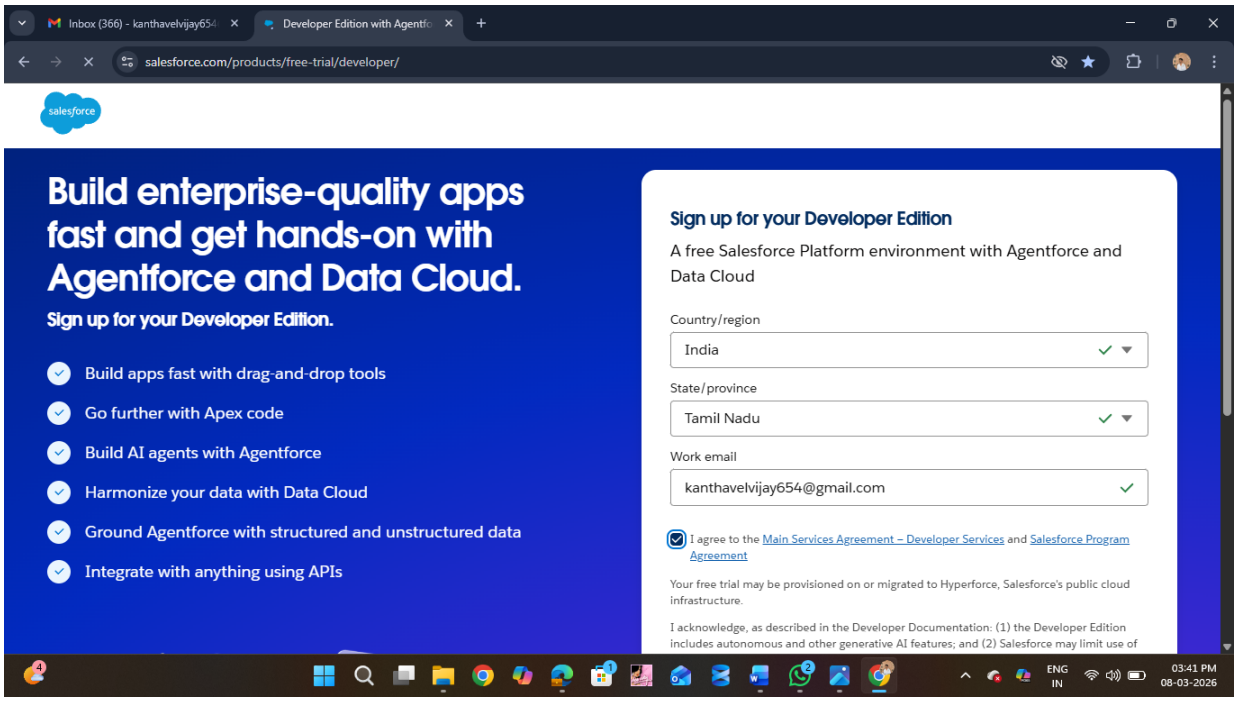
Team Creation

Creating a Salesforce Developer Edition

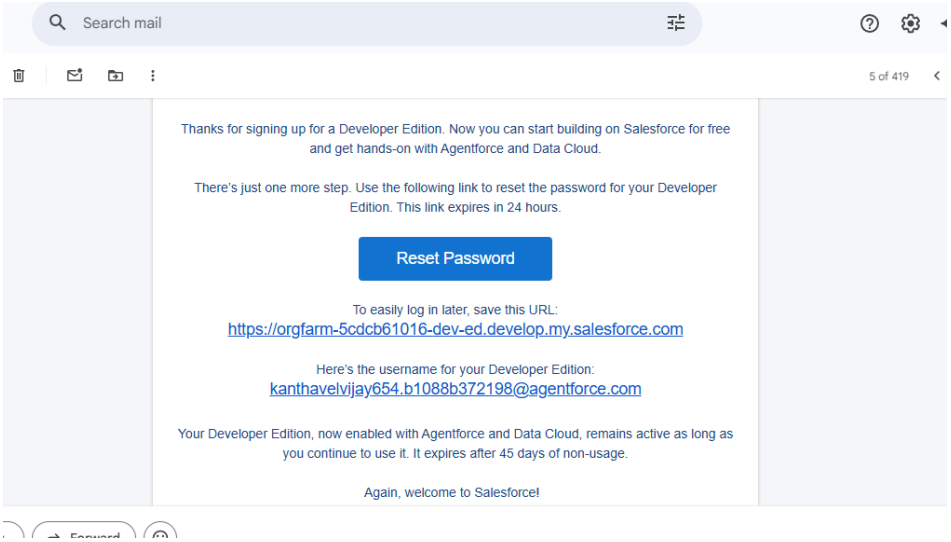
To develop the **CrickStats** application, a Salesforce Developer Edition must first be created. The team leader is responsible for creating the developer account and setting up the environment.

Steps to create the Developer Org:

1. Open the website: https://developer.salesforce.com/signup
2. Fill in the required details in the sign-up form:
    - First Name and Last Name
    - Email Address
    - Role: Developer
    - Company: College or Organization Name
    - Country: India
    - Postal Code: Your PIN Code
    - Username: Combination of your name and organization
3. After entering the details, click **Sign Me Up**.



4. A verification email will be sent to the registered email address. Open the email and click **Verify Account** to activate the Salesforce account. This email may take a few minutes to arrive.



5. Set a password and answer the security question, then click **Change Password**.

After completing these steps, the user will be redirected to the **Salesforce Setup page**, where the development of the **CrickStats application** can begin.

System Architecture

System Architecture

The **CrickStats** system is designed using two primary custom objects within Salesforce.

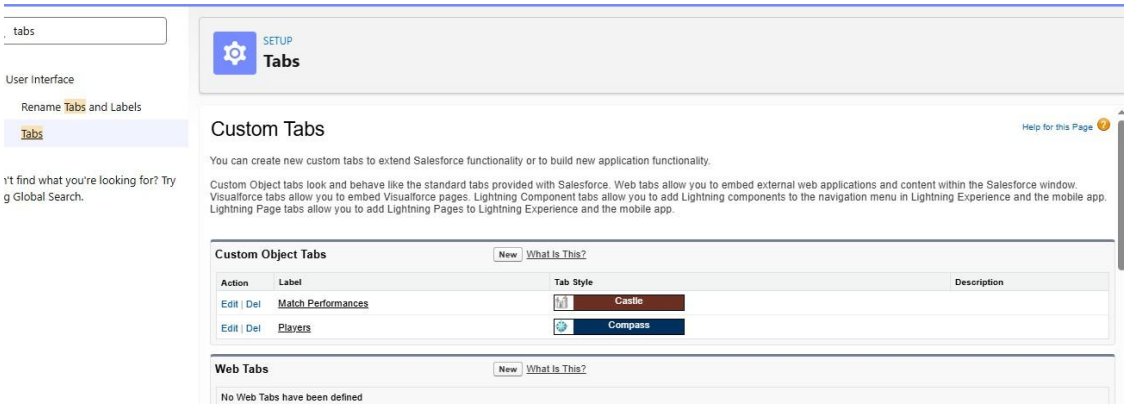
1. **Player**

   This object stores the player's personal information along with overall performance statistics collected from multiple matches.

2. **Match Performance**

   This object records detailed statistics for each match, including batting, bowling, and fielding performance.

A **Master–Detail Relationship** is established between the *Match Performance* object and the *Player* object. Through this relationship, roll-up summary fields can automatically calculate and display aggregated performance metrics for each player.



Player fields

SETUP > OBJECT MANAGER

Player

- Details
- Fields & Relationships
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters
- Search Layouts
- List View Button Layout
- Restriction Rules
- Scoping Rules
- Object Access
- Triggers
- Flow Triggers
- Validation Rules

Fields & Relationships

25 Items, Sorted by Field Label

FIELD LABEL	FIELD NAME	DATA TYPE
AVG Score	AVG_Score__c	Formula (Number)
Batting Style	Batting_Style__c	Picklist
Bowling Style	Bowling_Style__c	Picklist
Created By	CreatedById	Lookup(User)
Date of Birth	Date_of_Birth__c	Date
Last Modified By	LastModifiedById	Lookup(User)
Match Performance	Match_Performance__c	Lookup(Match Performance)
Matches Played	Matches_Played__c	Roll-Up Summary (COUNT Match Performance)
Nationality	Nationality__c	Picklist
Number of 100's	Number_of_100_s__c	Roll-Up Summary (COUNT Match Performance)
Number of 50's	Number_of_50_s__c	Roll-Up Summary (COUNT Match Performance)
Owner	OwnerId	Lookup(User,Group)
Player Name	Player_Name__c	Text(50)

- Details
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- Related Lookup Filters
- Search Layouts
- List View Button Layout
- Restriction Rules
- Scoping Rules
- Object Access
- Triggers
- Flow Triggers
- Validation Rules

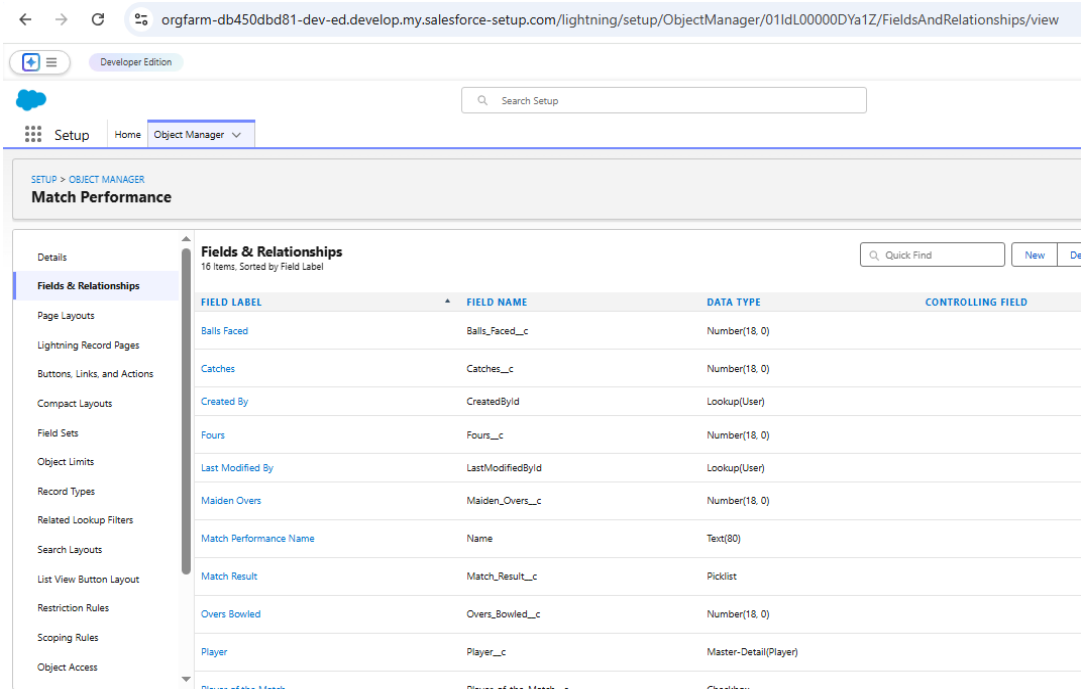
Fields & Relationships

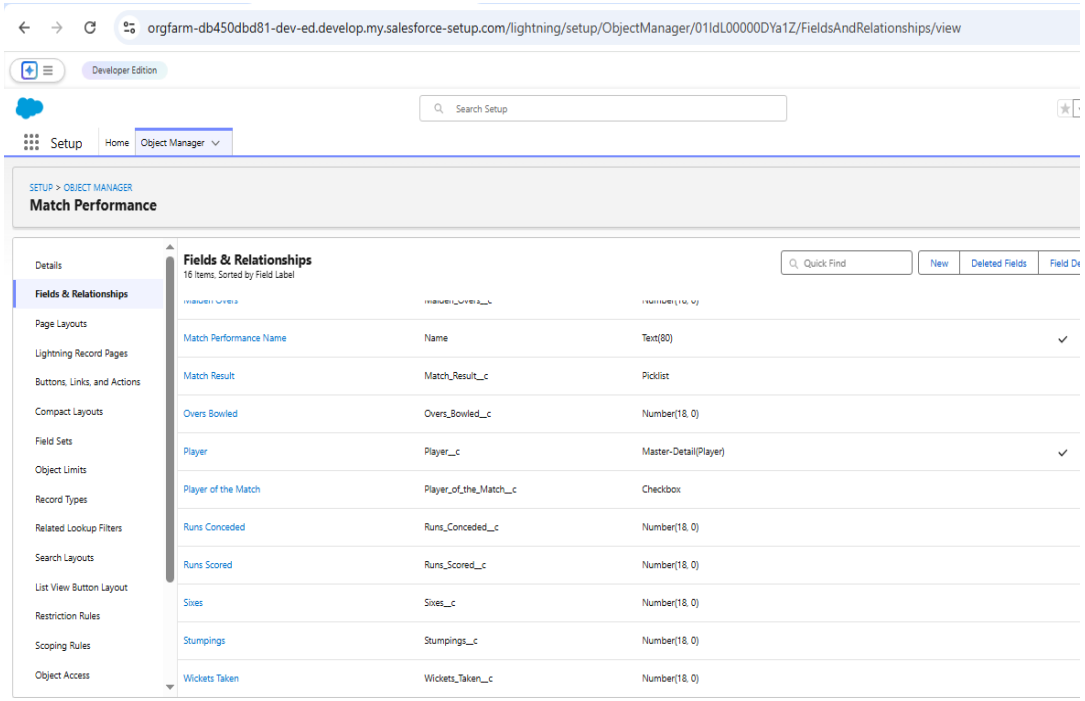
25 Items, Sorted by Field Label

Quick Find | New

Player Name	Player_Name__c	Text(50)
Player Name	Name	Text(80)
Player Number	Player_Number__c	Text(3)
Profile Picture	Profile_Picture__c	Rich Text Area(32768)
Role	Role__c	Picklist
Team	Team__c	Picklist
Total 4's	Total_4_s__c	Roll-Up Summary (SUM Match Performance)
Total 6's	Total_6_s__c	Roll-Up Summary (SUM Match Performance)
Total Catches	Total_Catches__c	Roll-Up Summary (SUM Match Performance)
Total Overs	Total_Overs__c	Roll-Up Summary (SUM Match Performance)
Total Runs	Total_Runs__c	Roll-Up Summary (SUM Match Performance)
Total Stumpings	Total_Stumpings__c	Roll-Up Summary (SUM Match Performance)
Total Wickets	Total_Wickets__c	Roll-Up Summary (SUM Match Performance)

Match Performance fields

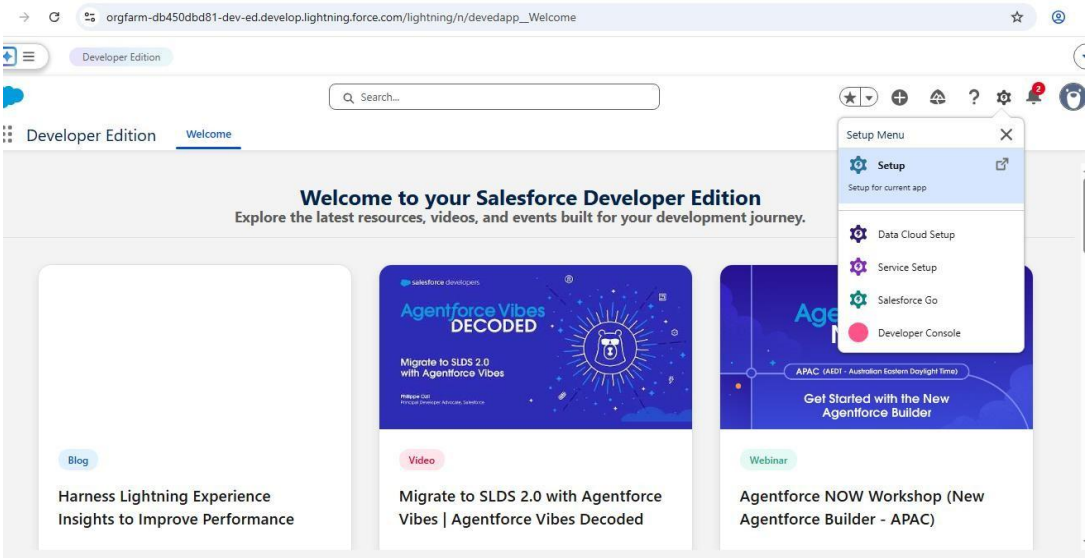




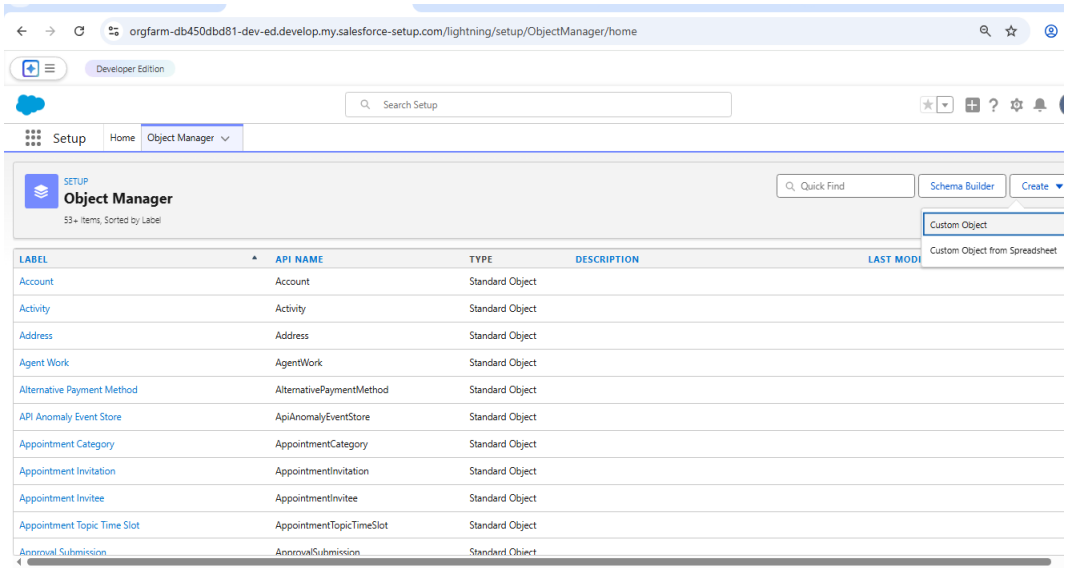
Data Model Design

Data Model Design

1. In your Salesforce org, click gear icon on the top left and select Setup to open Setup.



2. Click the Object Manager tab. If you don't see it, enter Object Manager in the Quick Find box.
3. On the Object Manager page, click Create | Custom Object



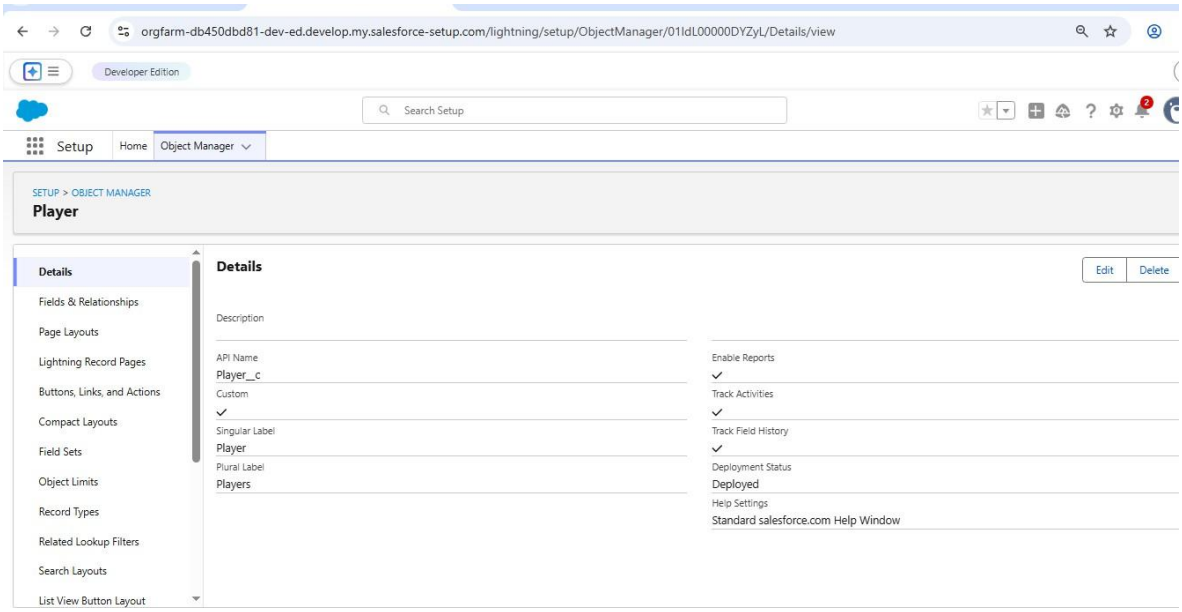
Player Object:

From the setup page → Click on Object Manager → Click on Create → Click on Custom Object.

- Label : Player
- Plural Label : Players
- Enter Record Name Label and Format

  Record Name : Player Name

  Data Type : Text
- In Optional features : check the boxes for Allow Reports | Allow Activities | Track Field History.
- Search Status : check the box for Allow Search.
- Save



The Player object stores personal information and aggregated statistics

Key fields include:

- Player Name
- Player Number
- Date of Birth
- Role
- Batting Style
- Bowling Style
- Nationality
- Team
- Profile Picture

Roll-up summary fields aggregate match statistics.

Examples:

- Matches Played
- Total Runs
- Total Wickets
- Total 4's
- Total 6's
- Total Overs
- Total Catches
- Total Stumpings

Match Performance Object:

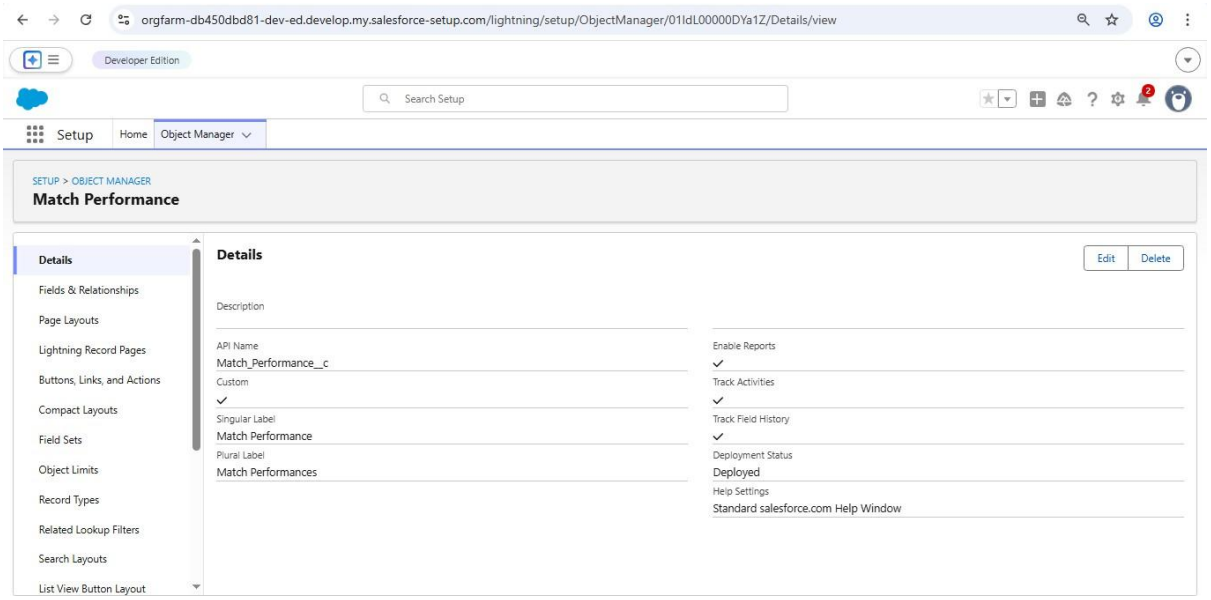
From the setup page → Click on Object Manager → Click on Create → Click on Custom Object.

- Label : Match Performance
- Plural Label : Match Performances
- Enter Record Name Label and Format

  Record Name : Customer Account Name

  Data Type : Text

- In Optional features : check the boxes for Allow Reports | Allow Activities | Track Field History.
- Search Status : check the box for Allow Search.
- Save



This object records match-specific statistics.

Key fields include:

- Runs Scored
- Balls Faced
- Fours
- Sixes
- Overs Bowled
- Wickets Taken
- Runs Conceded
- Maiden Overs
- Catches
- Stumpings

Formula fields calculate advanced metrics such as Strike Rate and Economy Rate.

Business Logic Implementation

Business Logic Implementation

The system uses formula fields and roll-up summary fields to automate cricket performance calculations.

STRIKE RATE FORMULA:

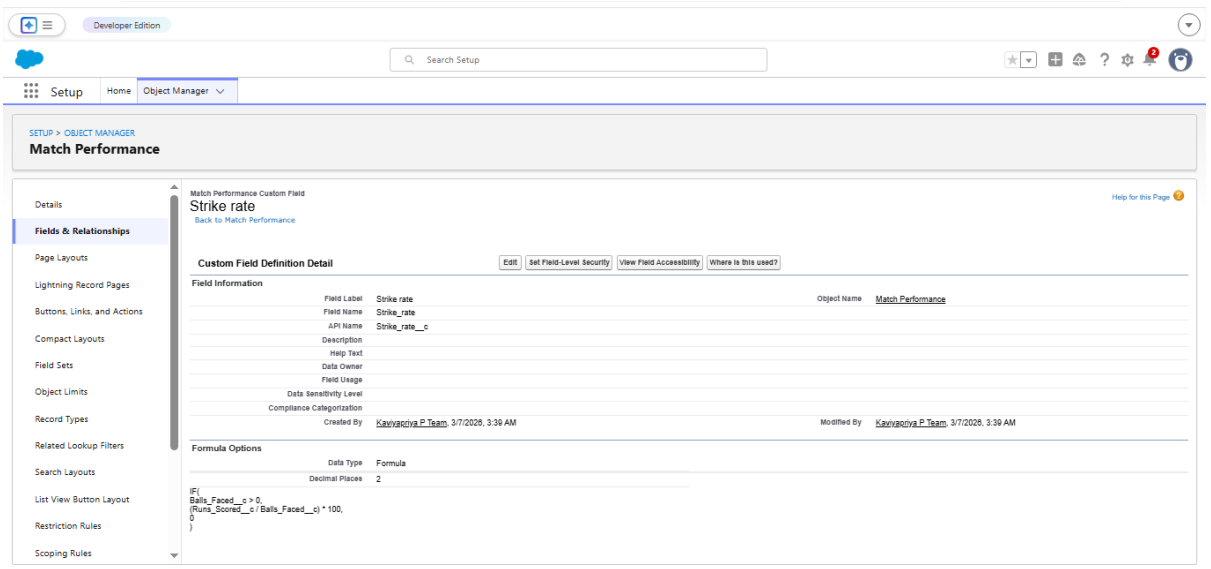
IF

(Balls_Faced__c > 0,

(Runs_Scored__c / Balls_Faced__c) * 100,

0)

This formula calculates the batting strike rate while preventing division-by-zero errors



ECONOMY RATE FORMULA:

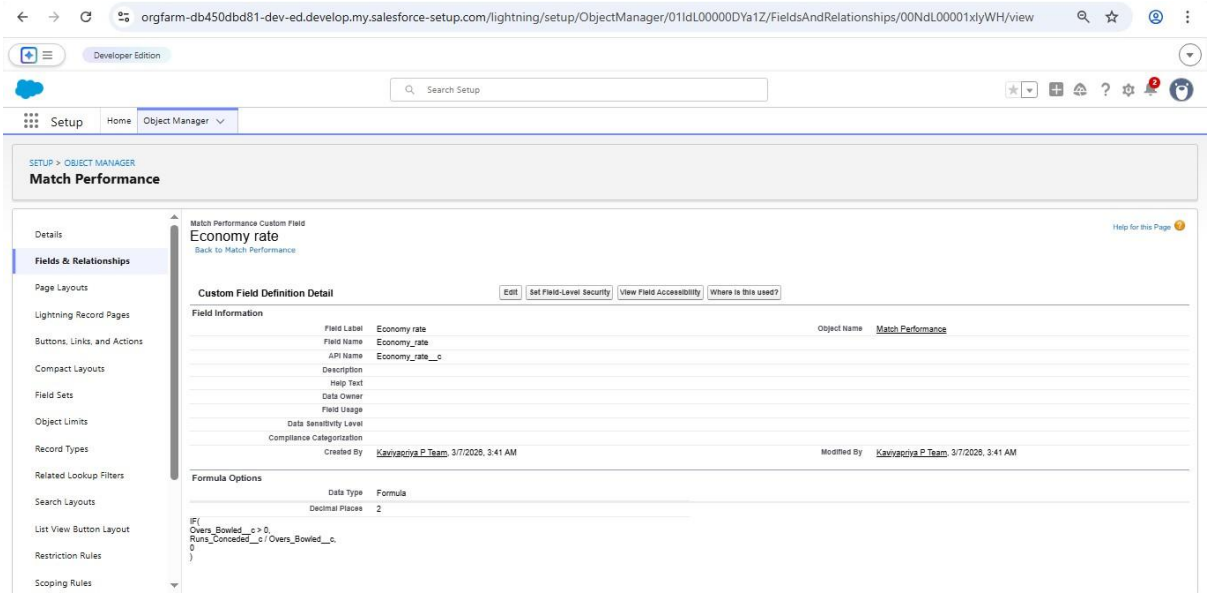
IF

(Overs_Bowled__c > 0,

Runs_Conceded__c / Overs_Bowled__c,

0)

This formula calculates the bowling economy rate



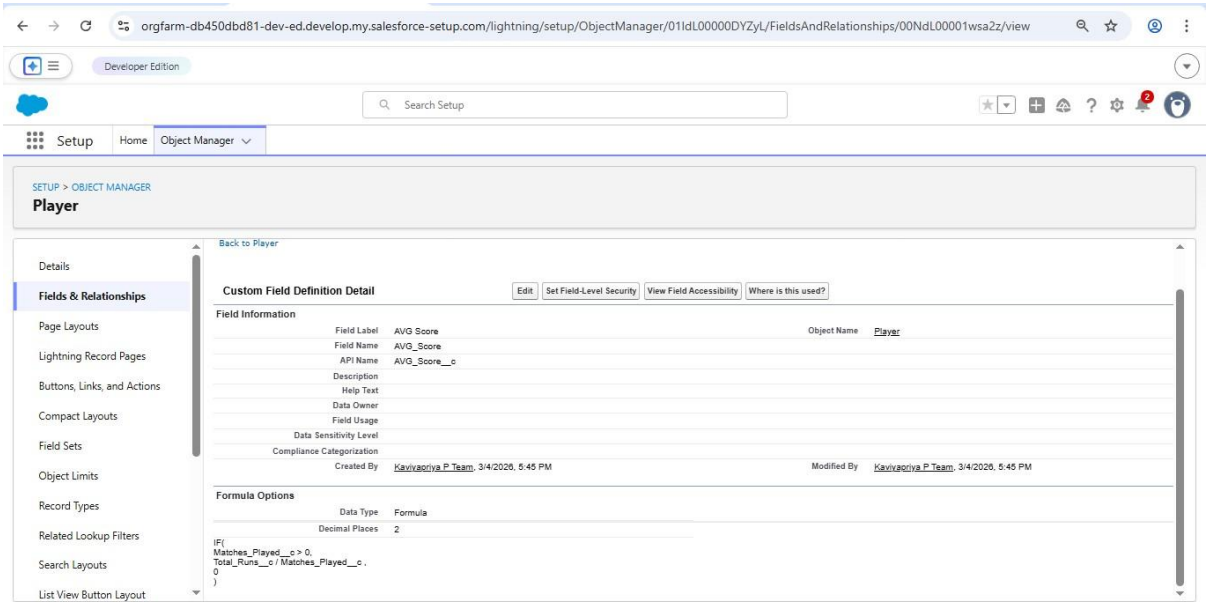
AVERAGE SCORE FORMULA:

IF(Matches_Played__c > 0,

Total_Runs__c / Matches_Played__c,

0)

This formula calculates the average runs scored per match.



Screen Flows

Screen Flows

Process of the Screen Flow:

1. **Start the Flow**

   The Screen Flow begins when the user opens the flow from the application interface. This acts as the starting point of the automated process.

2. **User Input Screen**

   A screen component is displayed where the user is asked to select a **team name from a picklist**. This allows the user to choose the team whose players they want to view.

3. **Get Records Element**

   After the user selects the team, the flow uses a **Get Records** element to retrieve player details from the **Player object**. The system filters the records based on the selected team.

4. **Store Retrieved Data**

   The player records that match the selected team are stored temporarily within the flow for further processing.

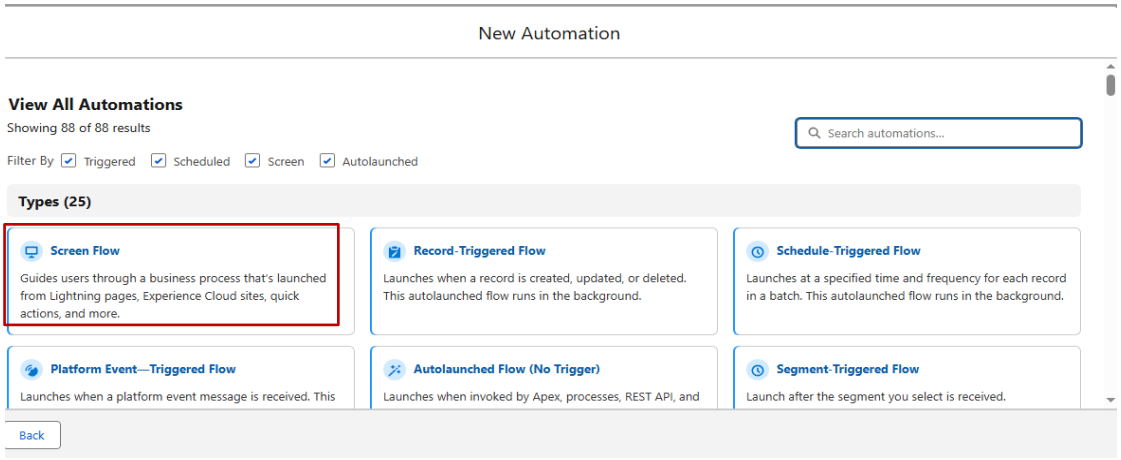
5. **Display Results**

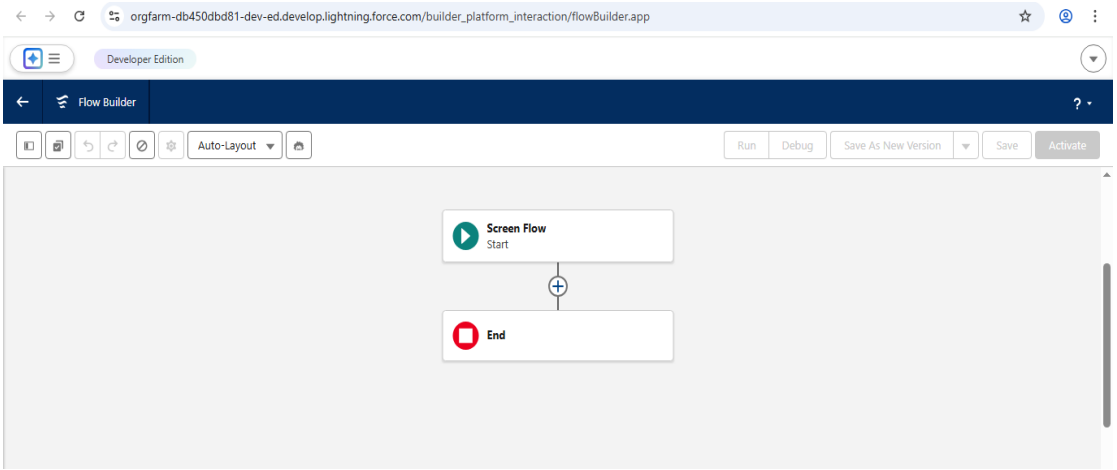
   The filtered player data is then displayed on another screen using a **Data Table component**, showing the list of players belonging to the selected team.

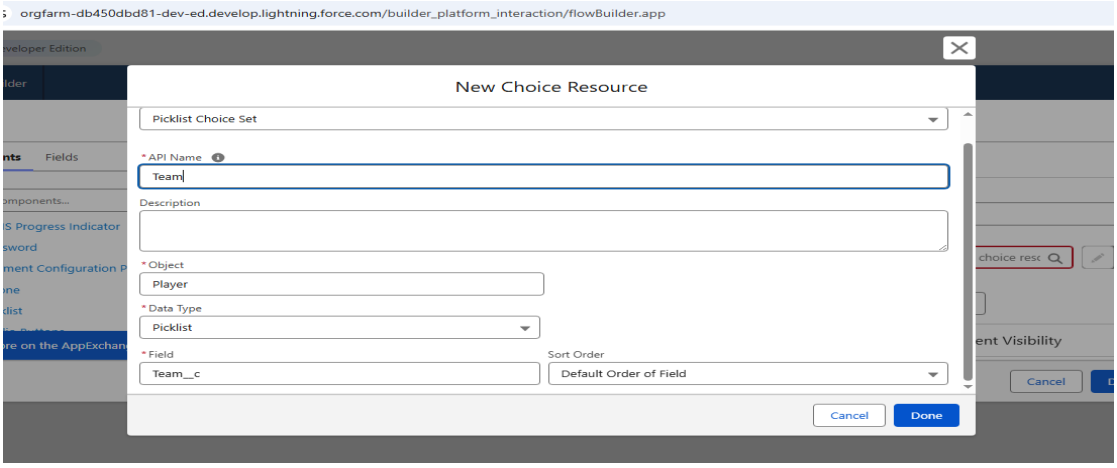
6. **End of Flow**

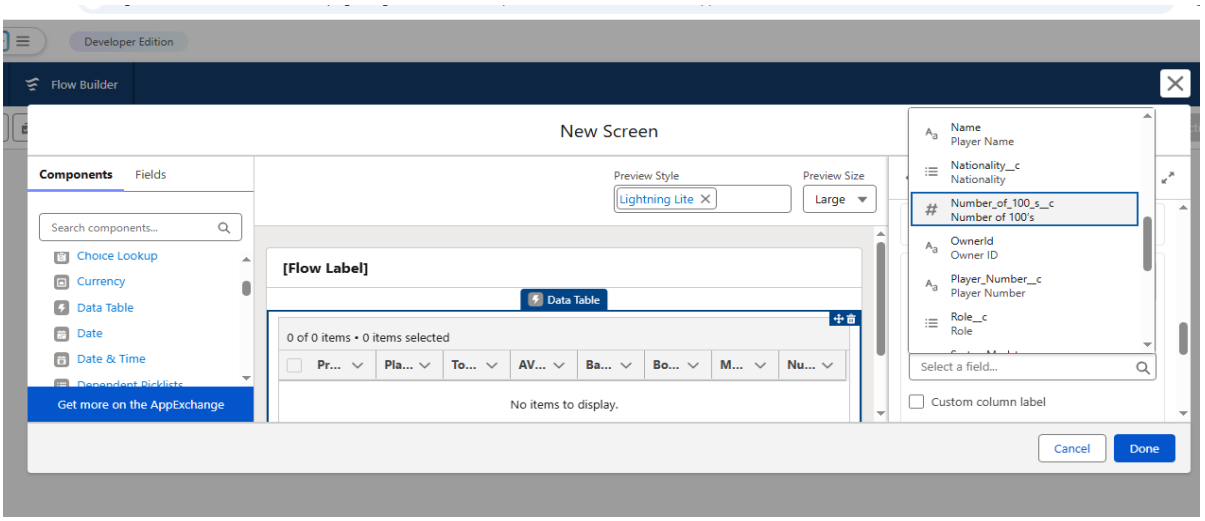
   After displaying the results, the flow process is completed. The user can view the filtered player list and exit the flow.

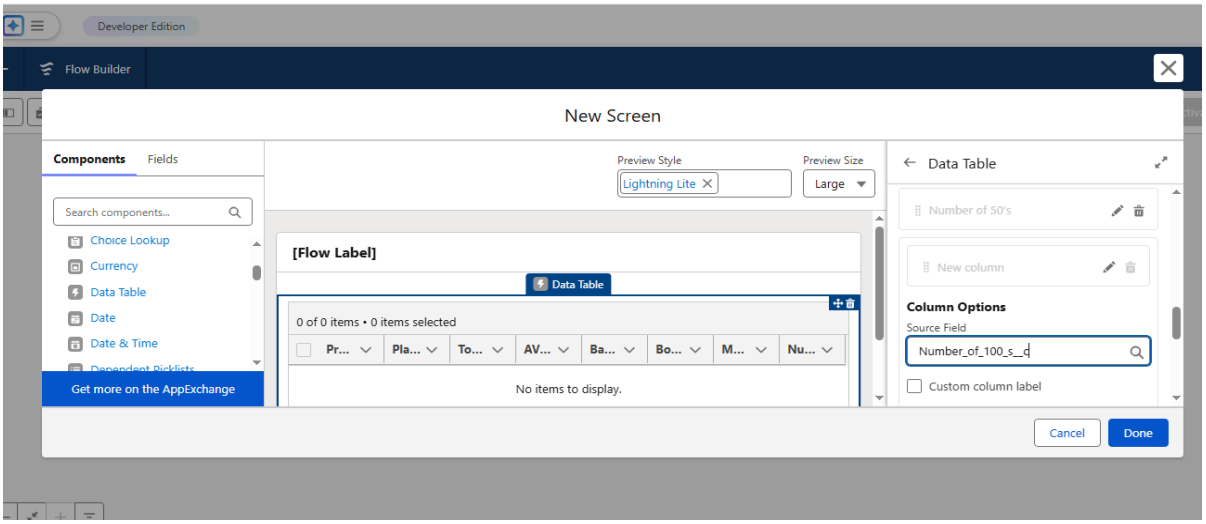
Screenshots of Screen flows








Developer Edition
Flow Builder
New Screen
Components
Fields
Search components...
Choice Lookup
Currency
Data Table
Date
Date & Time
Get more on the AppExchange
Preview Style
Lightning Lite
Preview Size
Large
[Flow Label]
Data Table
0 of 0 items • 0 items selected
No items to display.
Name
Player Name
Nationality__c
Nationality
Number_of_100_s__c
Number of 100's
OwnerId
Owner ID
Player_Number__c
Player Number
Role__c
Role
Select a field...
Custom column label
Cancel
Done


Developer Edition
Flow Builder
New Screen
Components
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Search components...
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Get more on the AppExchange
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Lightning Lite
Preview Size
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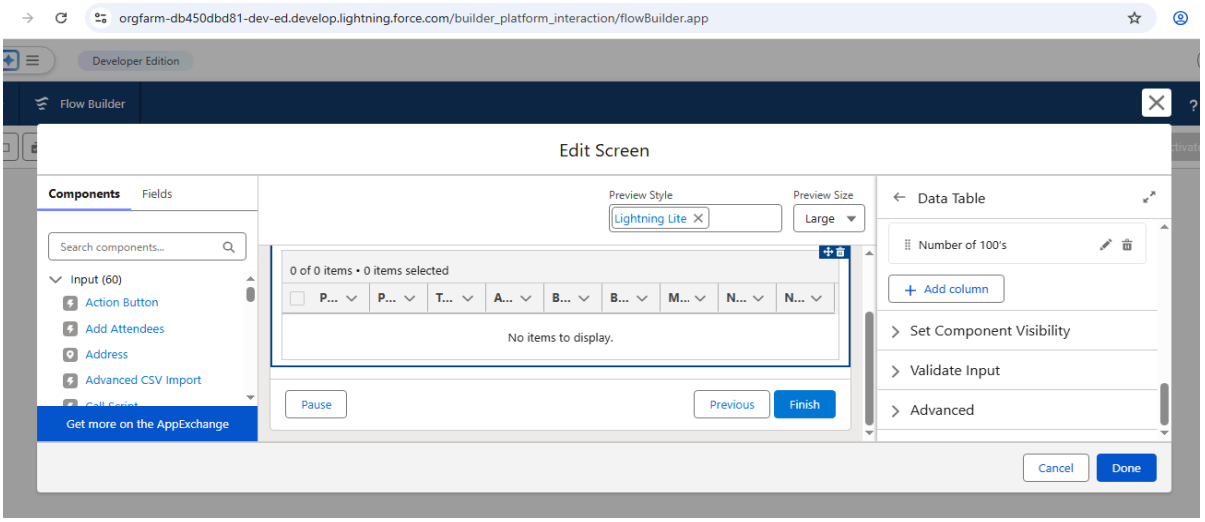
Data Table
0 of 0 items • 0 items selected
No items to display.
Data Table
Number of 50's
New column
Column Options
Source Field
Number_of_100_s__c
Custom column label
Cancel
Done


orgfarm-db450dbd81-dev-ed.develop.lightning.force.com/builder_platform_interaction/flowBuilder.app
Developer Edition
Flow Builder
Edit Screen
Components
Fields
Search components...
Input (60)
Action Button
Add Attendees
Address
Advanced CSV Import
Get more on the AppExchange
Preview Style
Lightning Lite
Preview Size
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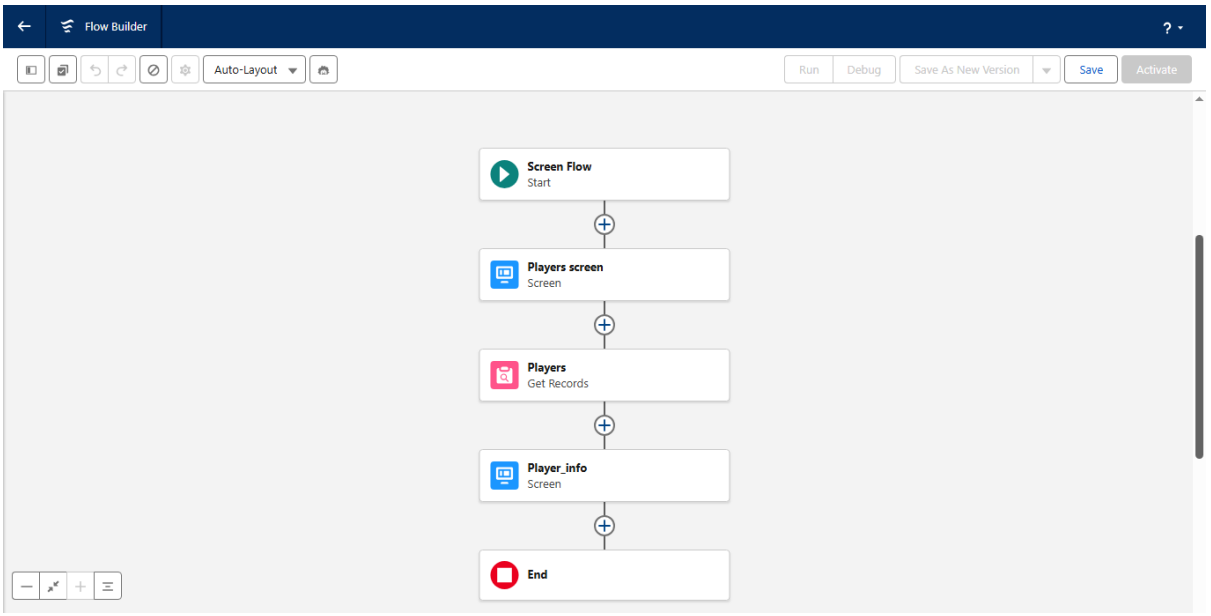
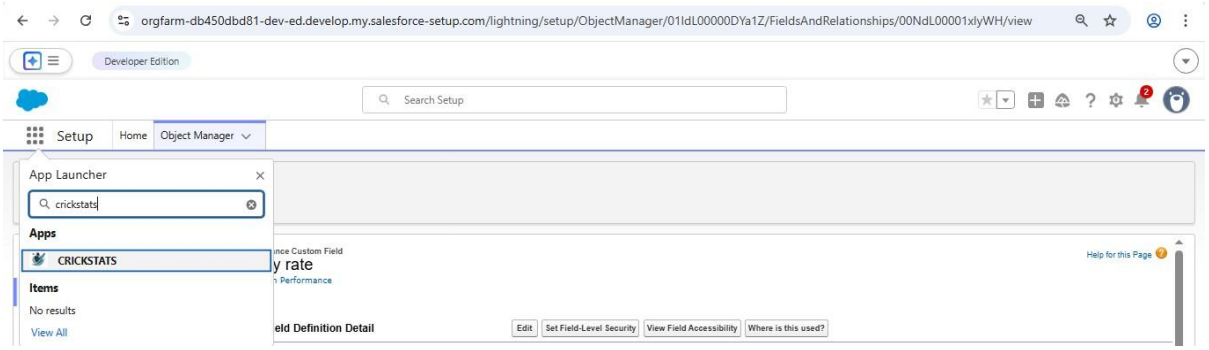
0 of 0 items • 0 items selected
No items to display.
Pause
Previous
Finish
Data Table
Number of 100's
Add column
Set Component Visibility
Validate Input
Advanced
Cancel
Done


Flow Builder
Auto-Layout
Run
Debug
Save As New Version
Save
Activate
Screen Flow
Start
Players screen
Screen
Players
Get Records
Player_info
Screen
End

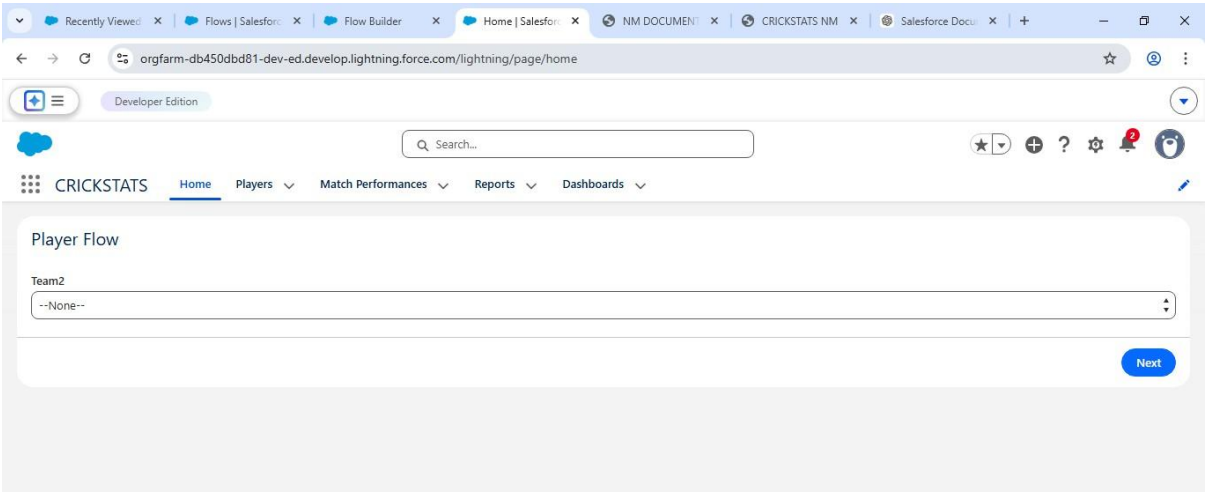
Flow Automation

Flow Automation

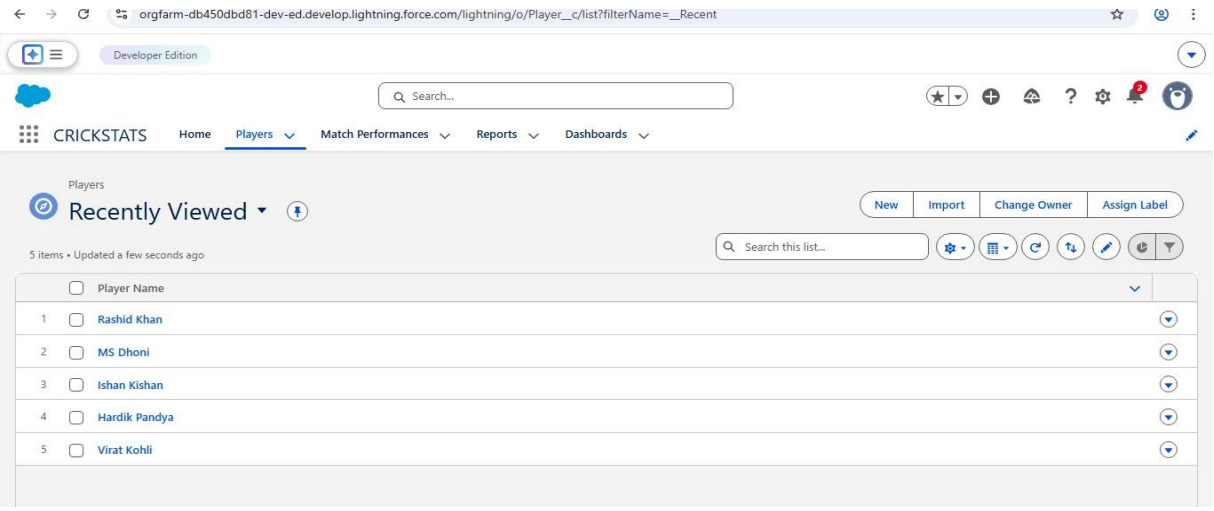
1. In the **CrickStats** application, users can open the app through the **App Launcher** by searching and selecting the CrickStats application.



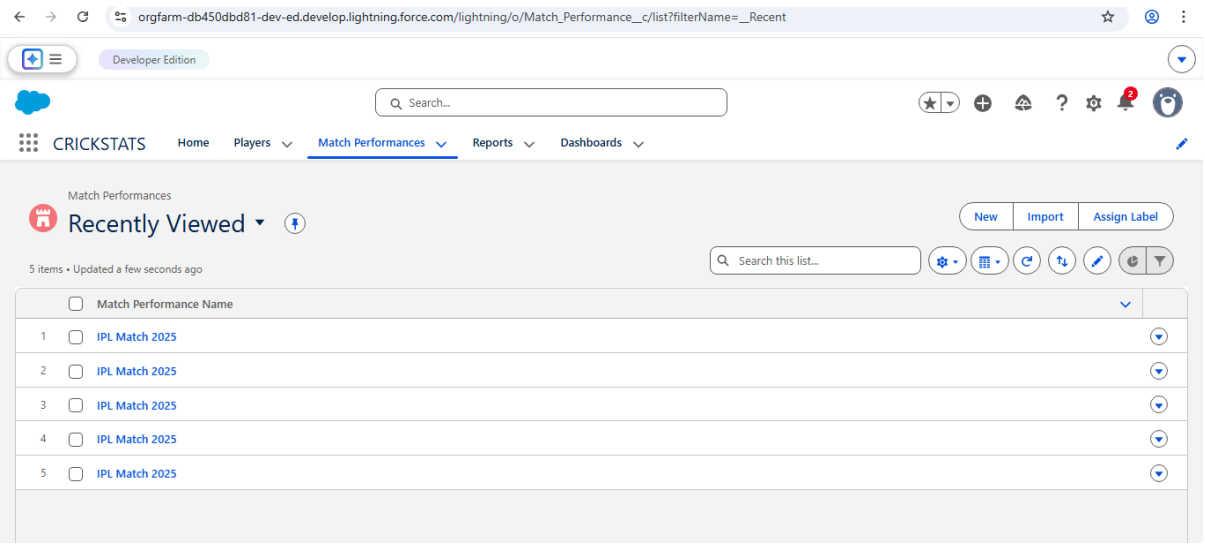
2. After opening the application, several tabs are available such as **Home, Players, Match Performance, Reports, and Dashboards**.



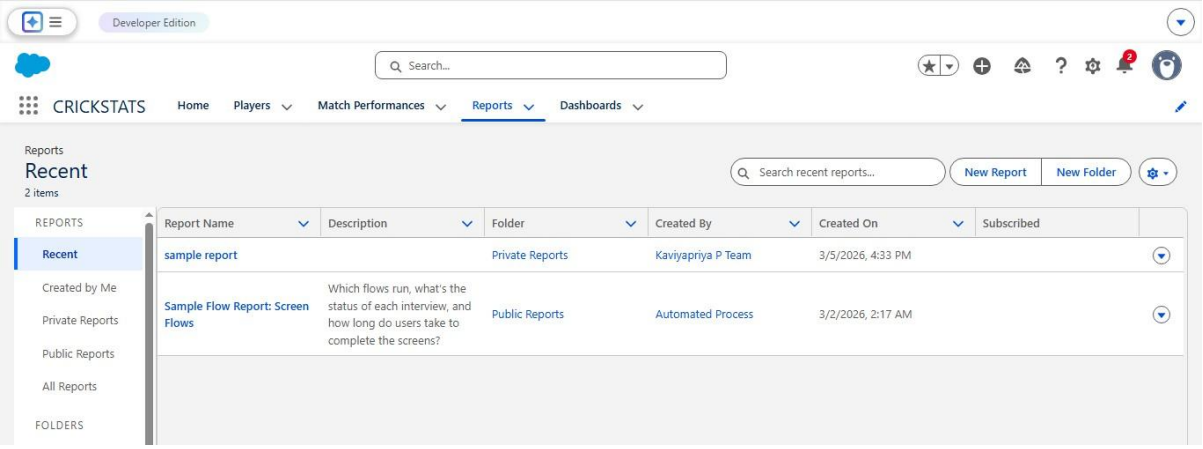
3. The **Players tab** is used to store player details such as player name, number, role, batting style, bowling style, and other personal information.



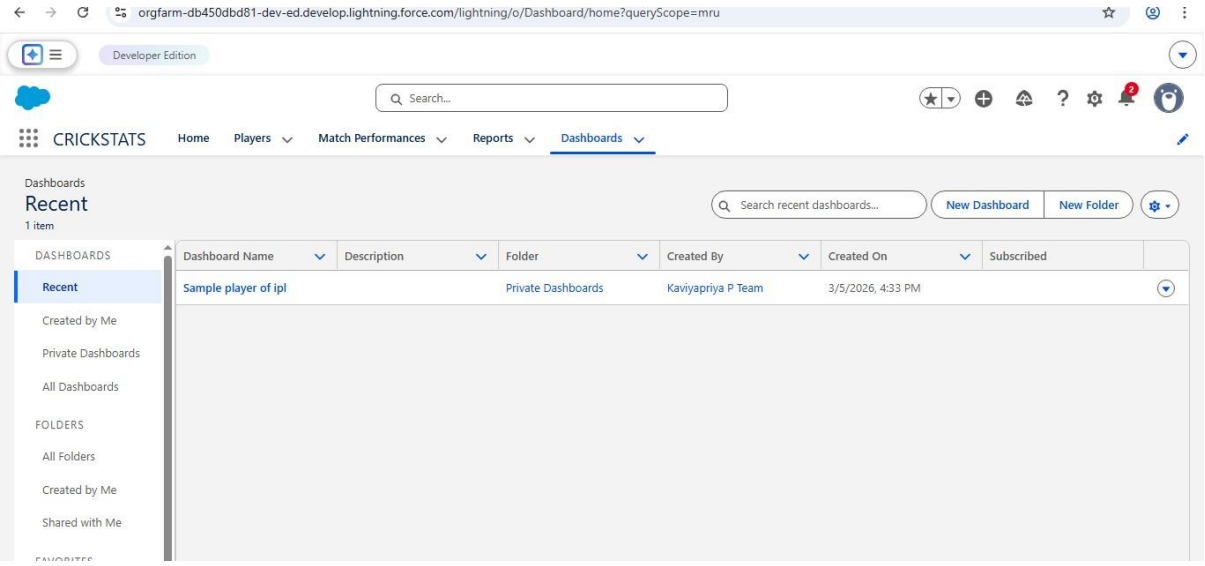
4. The **Match Performance tab** records match-by-match statistics including runs scored, balls faced, fours, sixes, overs bowled, wickets taken, runs conceded, catches, and stumpings.



5. The **Reports tab** is used to generate analytical reports such as top run scorers, top wicket takers, and overall player performance rankings.



6. The **Dashboards tab** visually displays player statistics using charts, graphs, and tables to help users quickly understand performance insights.

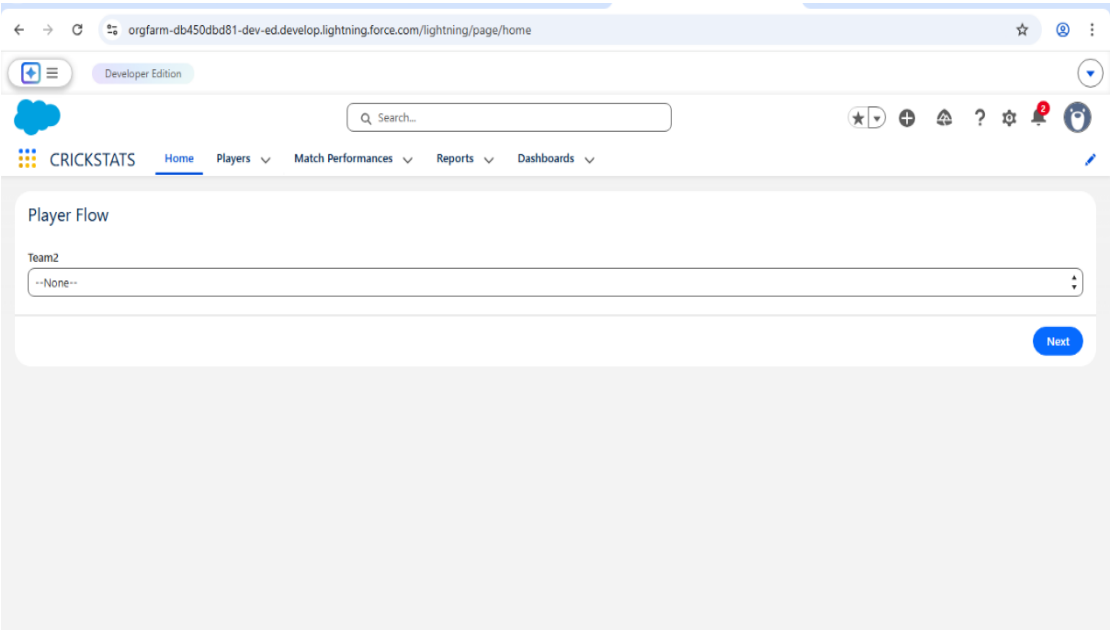


7. Flow automation helps users interact with the system efficiently and makes it easier to retrieve and analyze cricket performance data within the CrickStats application.

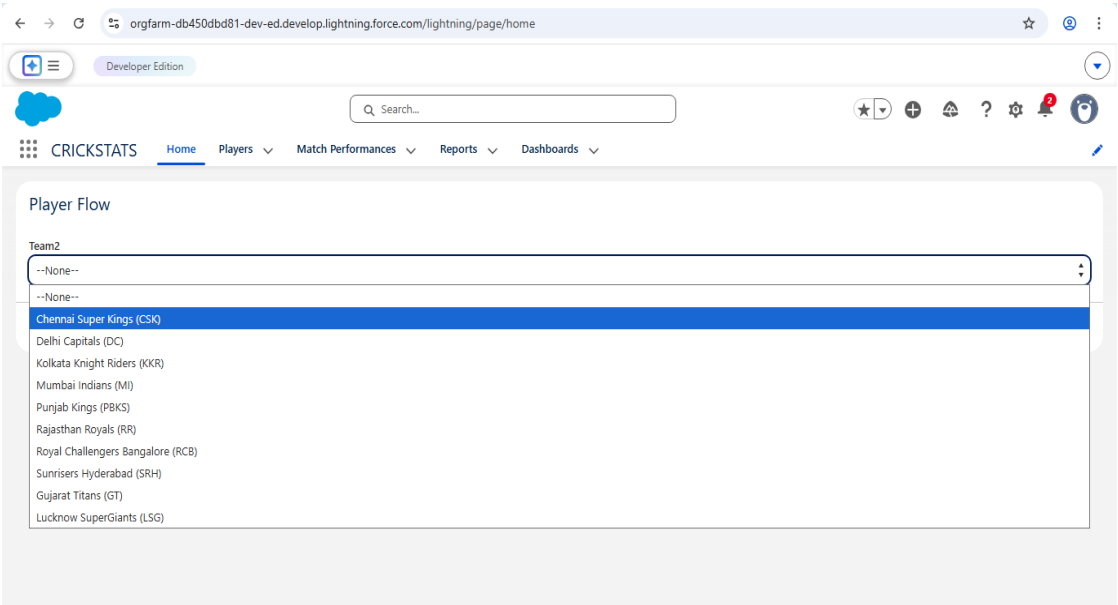
Home

Home

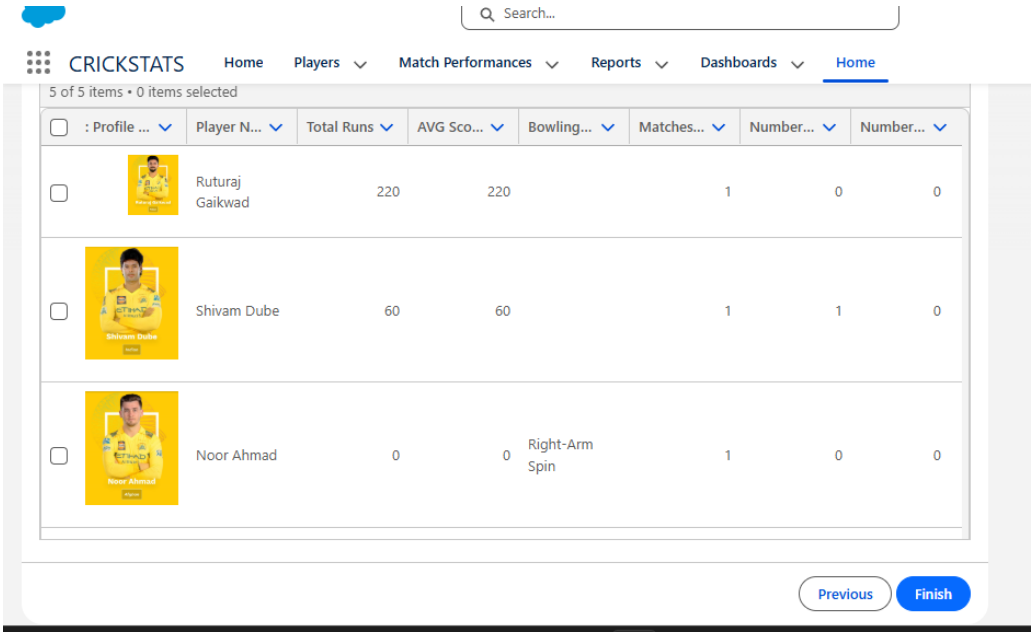
- In the **Home tab** of the CrickStats application, an **item bar with a dropdown list** is available that displays all **IPL teams**.



- The user can **select any team** from the dropdown list.



- After selecting the team and clicking **Next**, the system displays the **players registered under that selected team**.

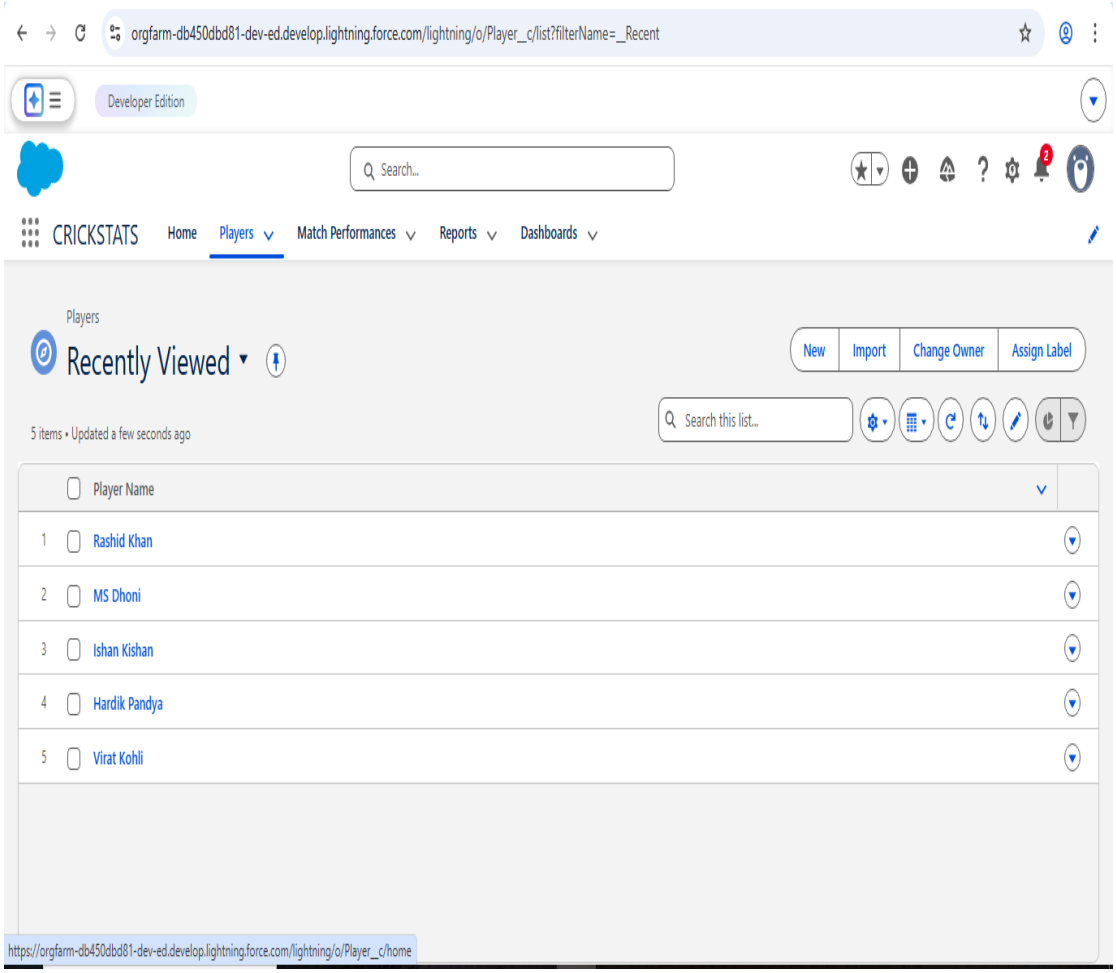


- The system shows the **complete details of the players**, including their information from the **Player list and Match Performance records**.
- This allows users to **view player statistics and performance details of a particular team easily**.

Players

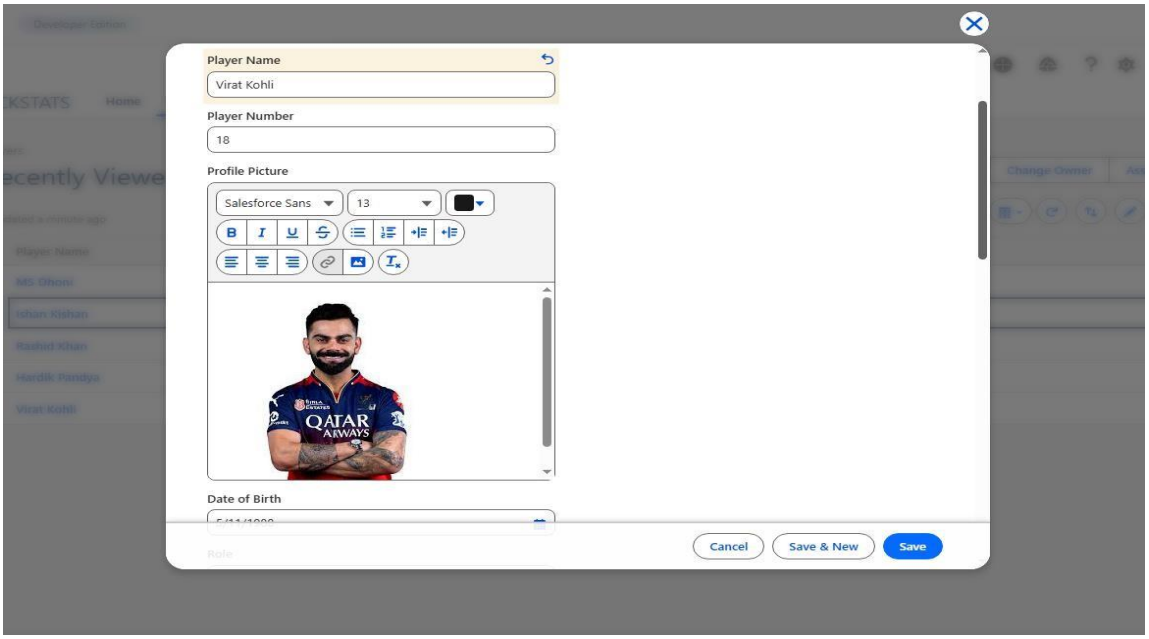
Players

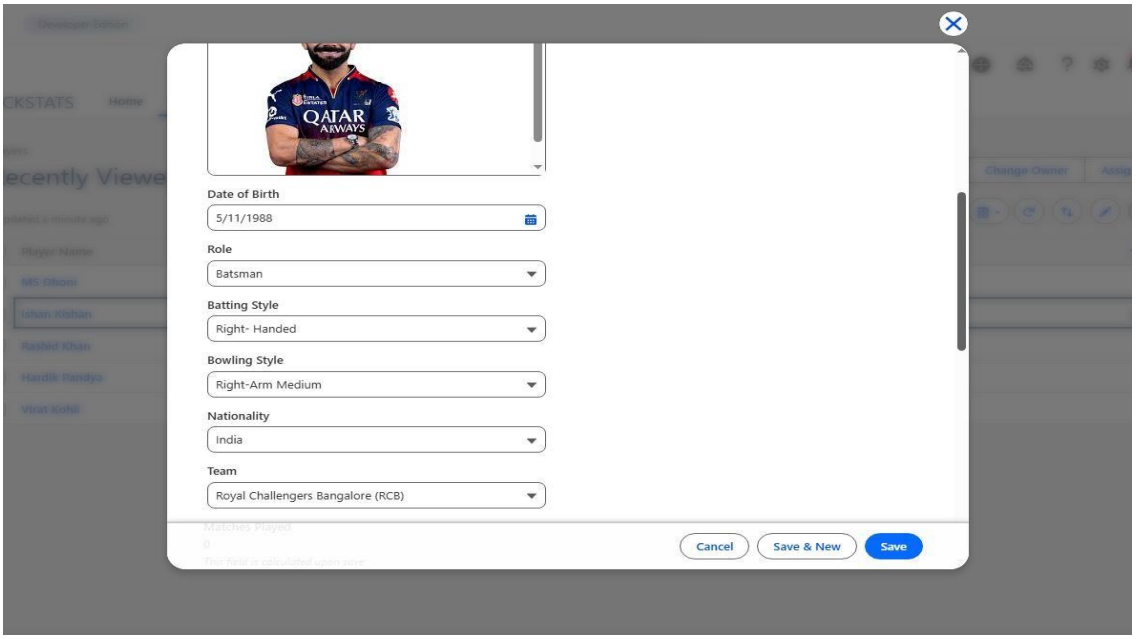
- In the **Players tab**, the system initially displays the list of all players who are already registered in the application.

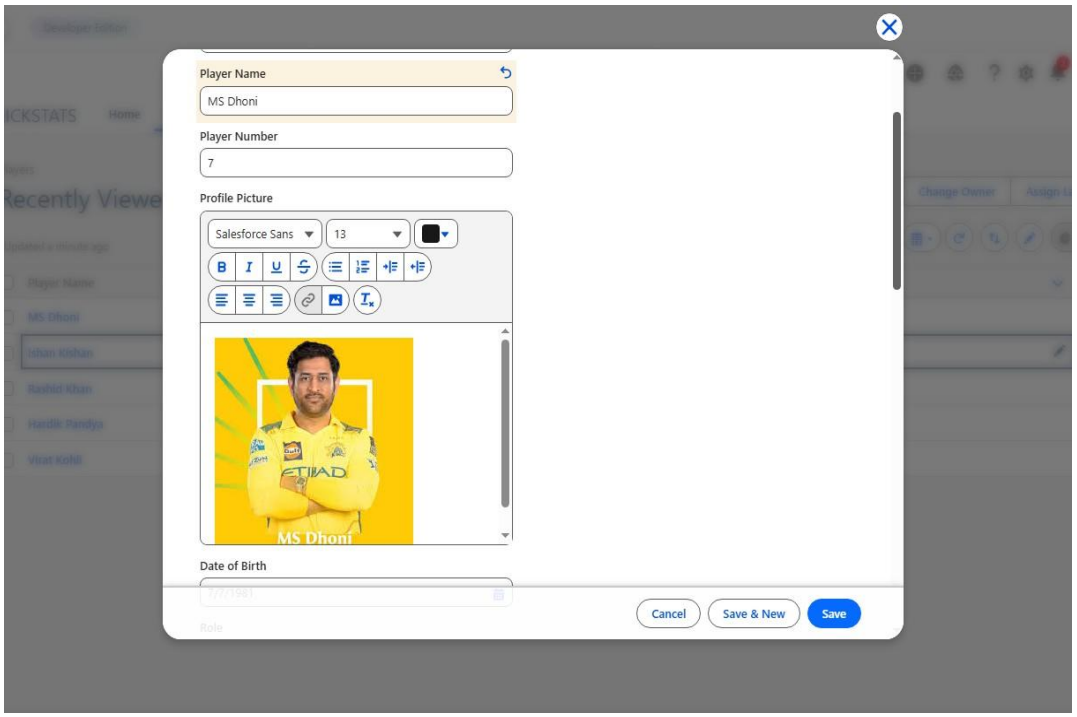


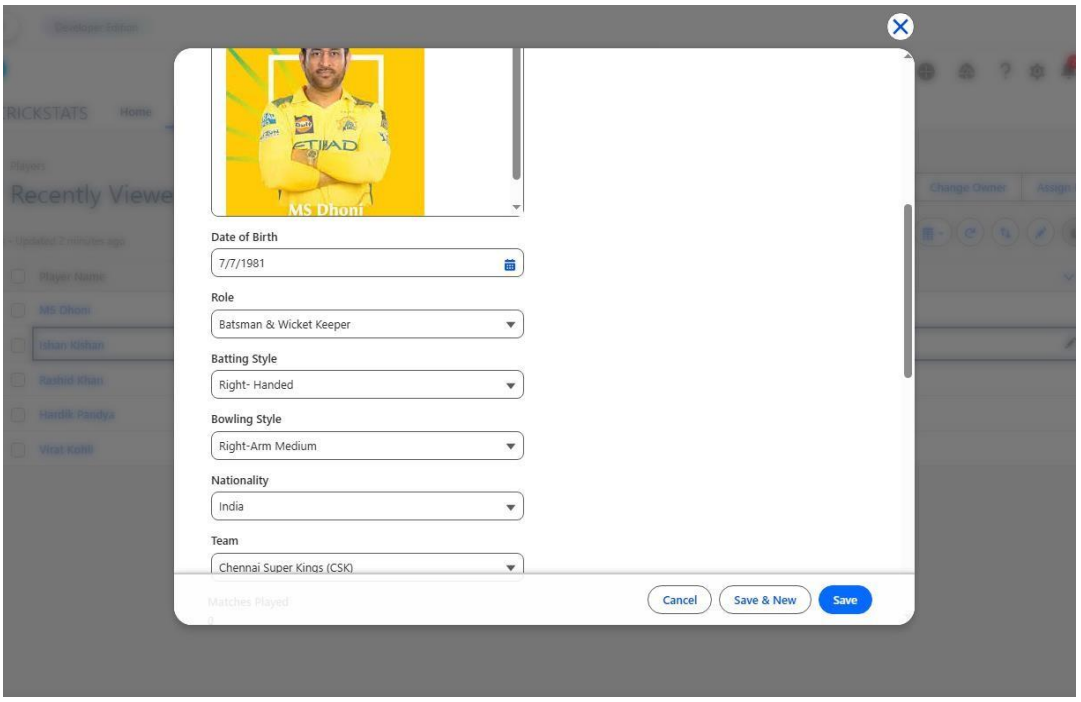
- Users can **view, edit, or delete** the existing player records from this list.

- The tab also provides an option to **add new players** by entering their details.

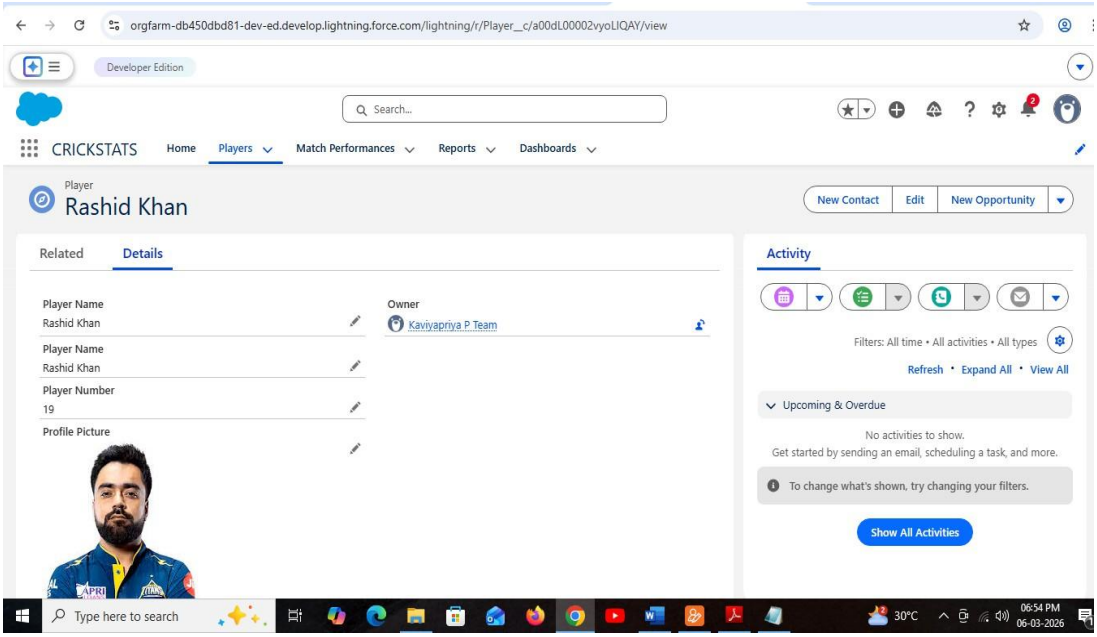


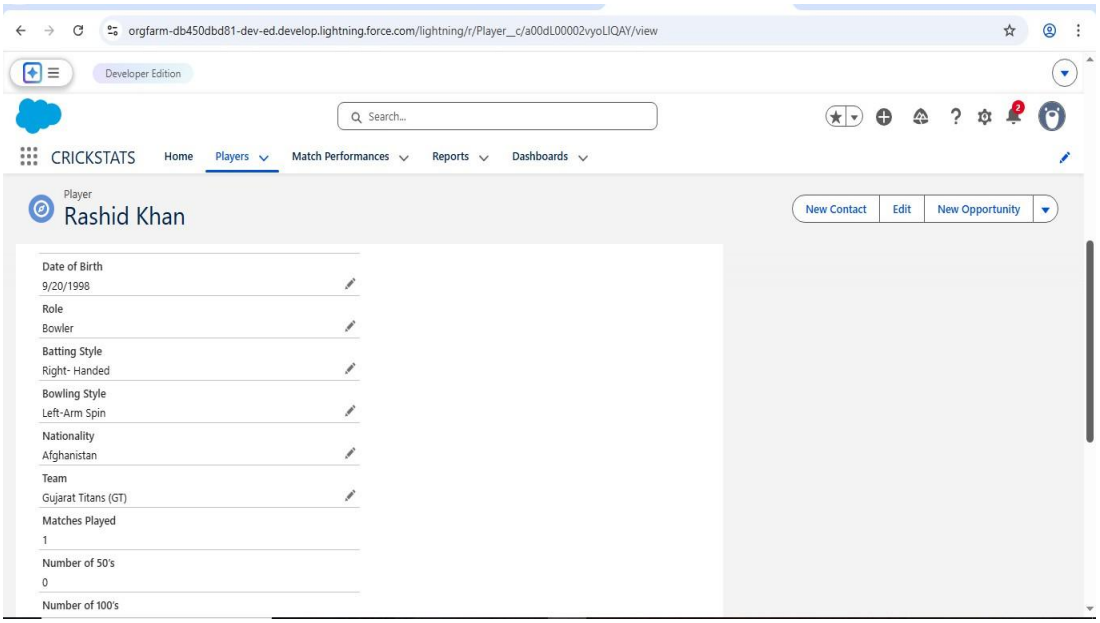


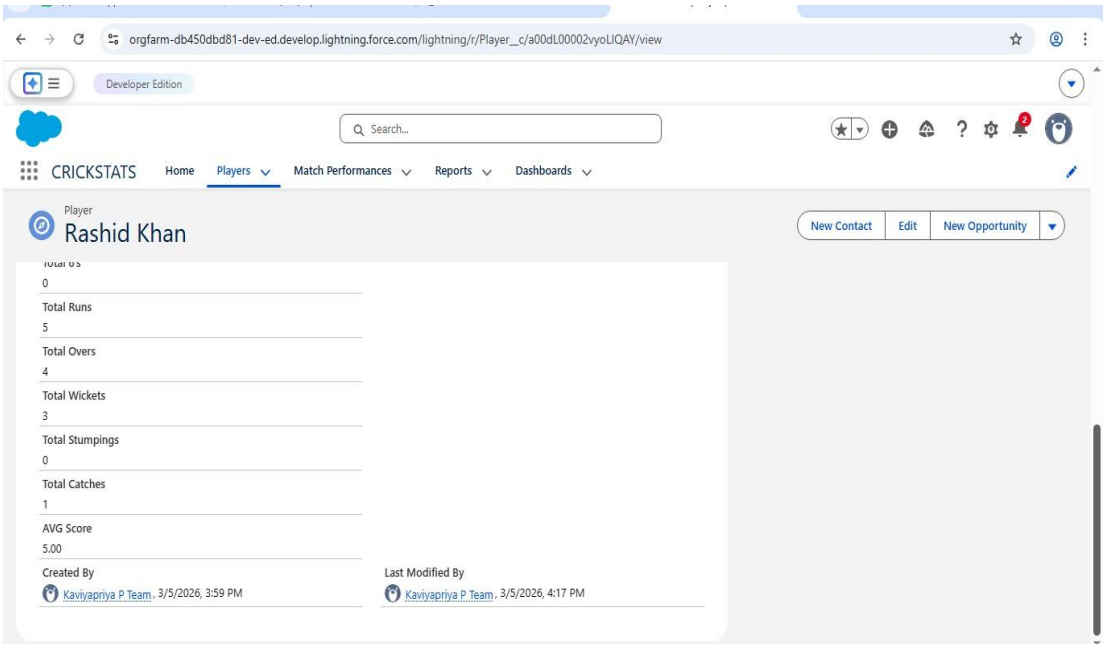

Player Name
MS Dhoni
Player Number
7
Profile Picture
Salesforce Sans
13
MS Dhoni
Date of Birth
Cancel
Save & New
Save


MS Dhoni
Date of Birth
7/7/1981
Role
Batsman & Wicket Keeper
Batting Style
Right- Handed
Bowling Style
Right-Arm Medium
Nationality
India
Team
Chennai Super Kings (CSK)
Cancel
Save & New
Save

- When a user clicks on a **specific player's name**, the system opens that player's **profile page**.
- The profile page shows the **personal information of the player** along with their **match performance details**, including the statistics recorded for each match.





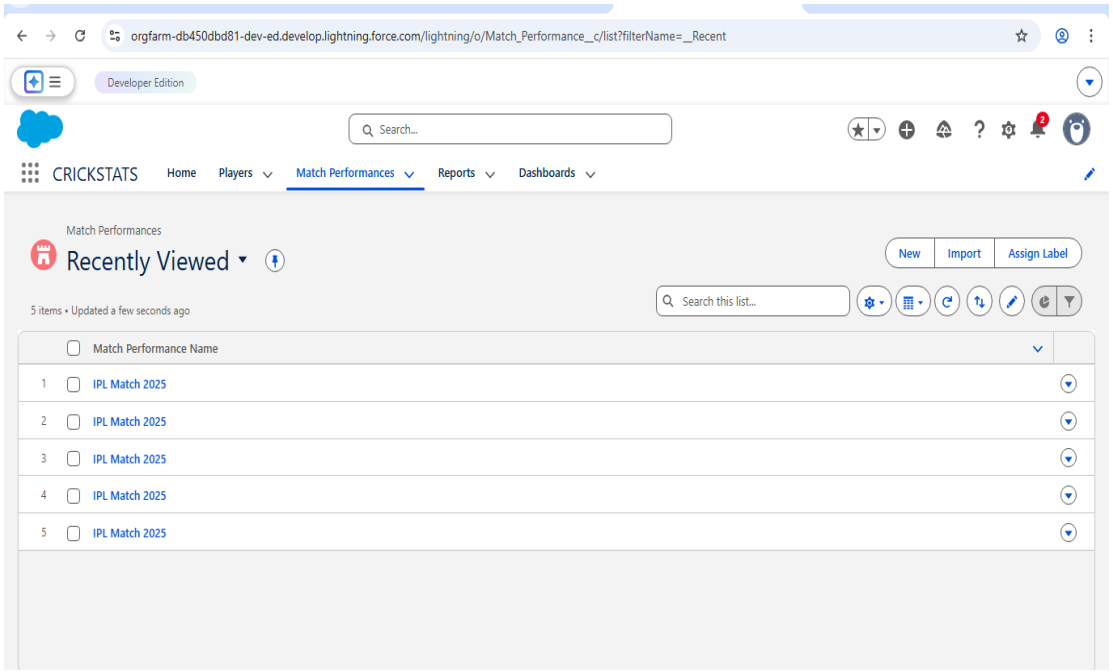


- All the relevant information related to the player is displayed clearly, allowing users to **check and verify the player's complete performance data**.

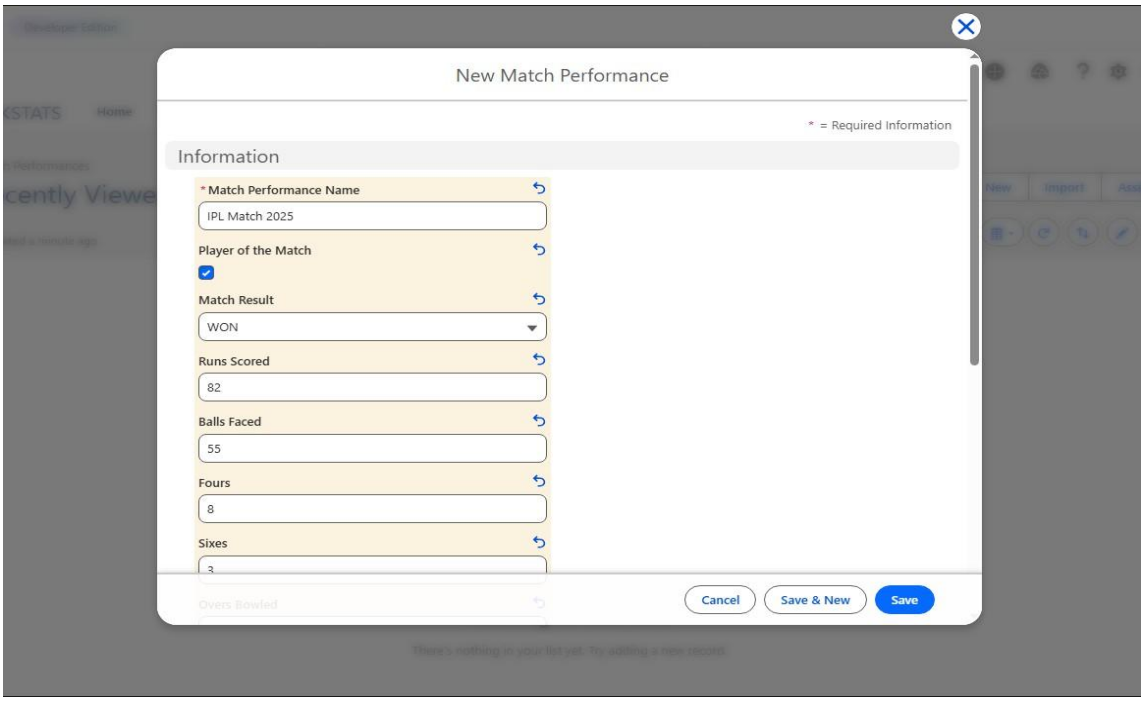
Match Performances

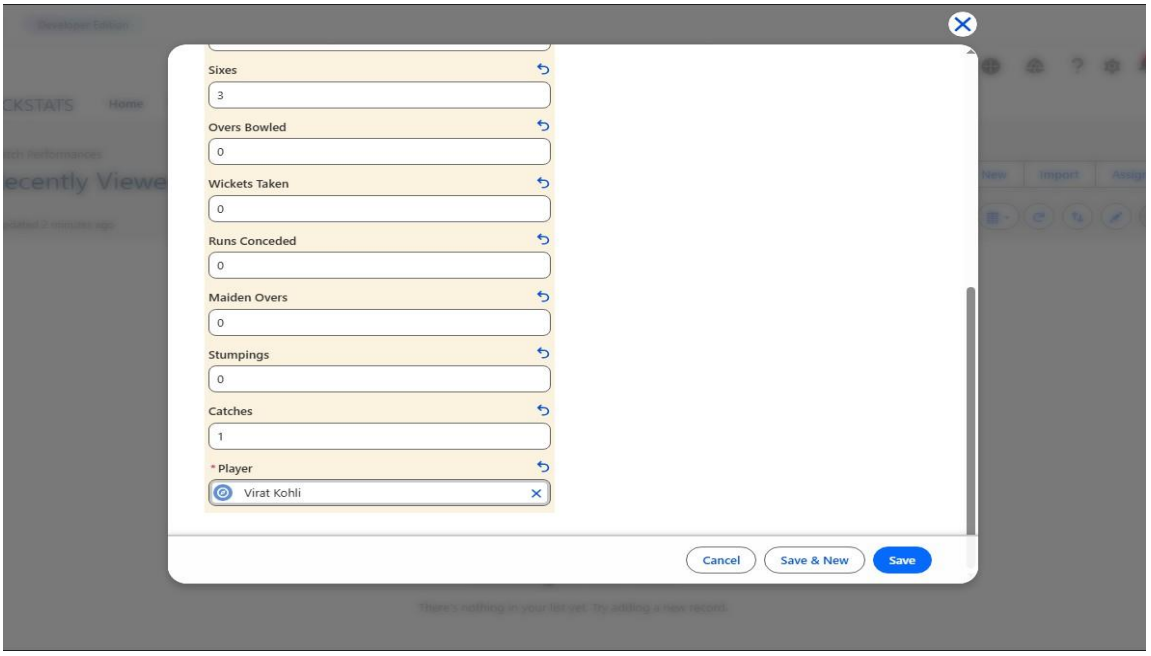
Match Performances

- In the **Match Performance tab**, the system displays the list of matches along with the **year in which the matches were played**.

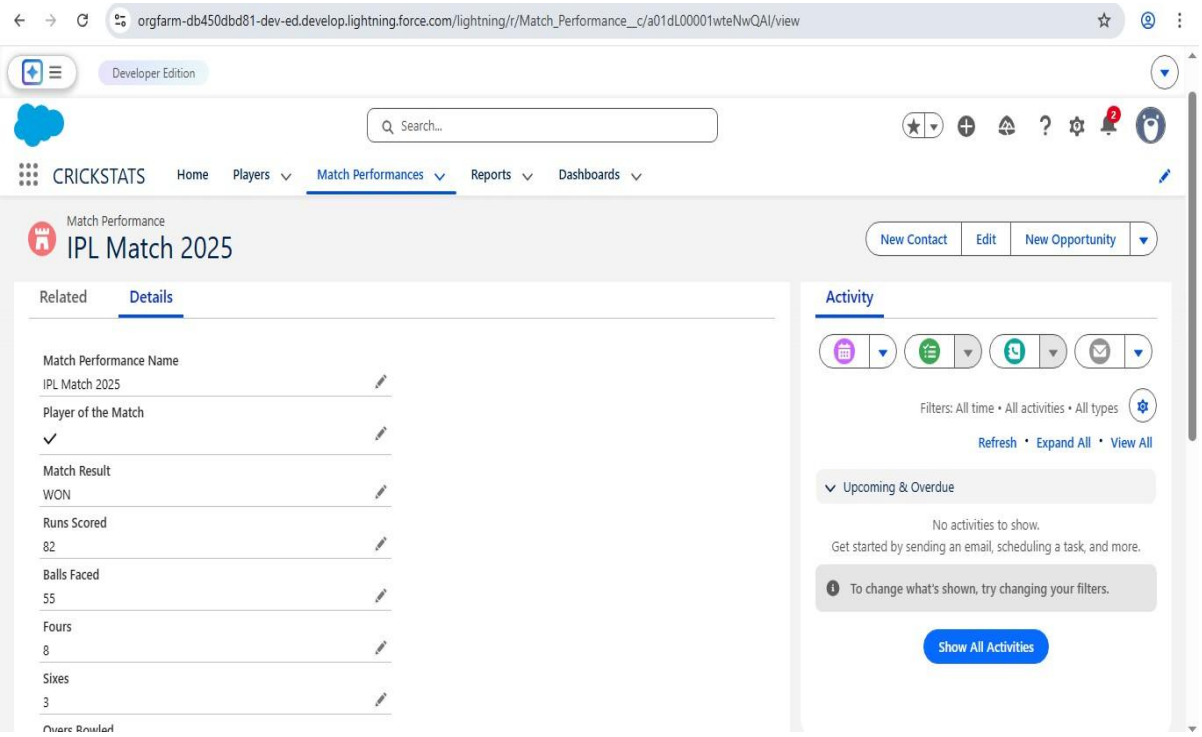


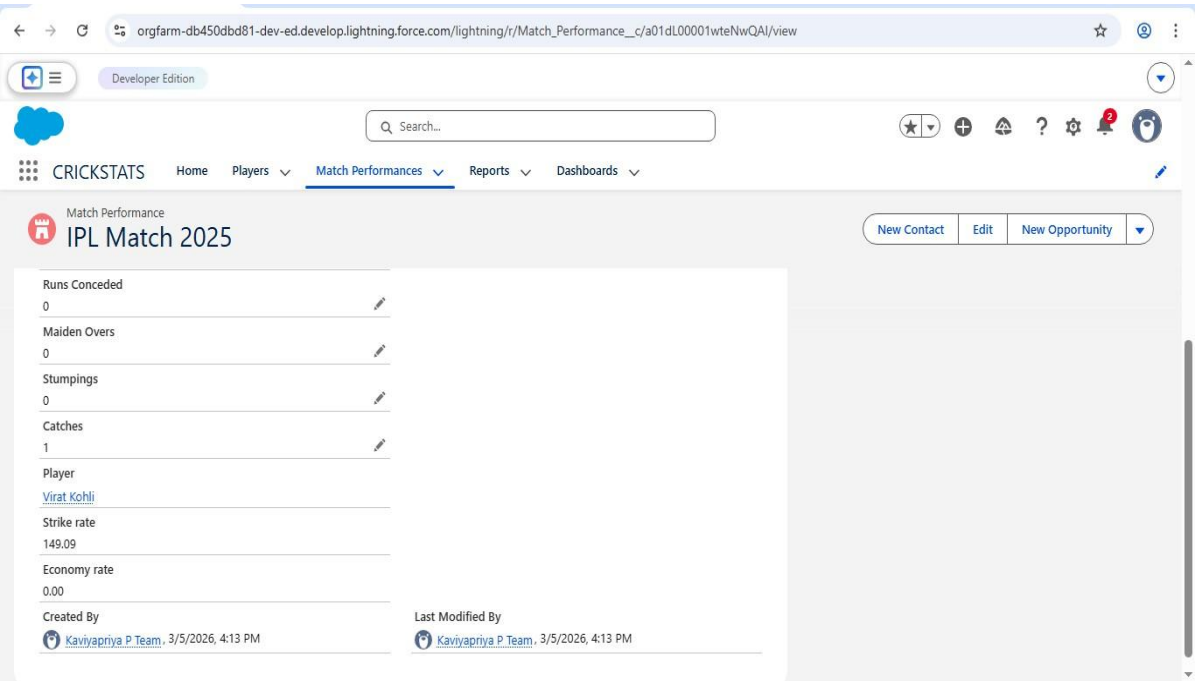
- Match performance records can be created **only for players who are already registered in the Players tab**. The system does not allow creating match records for players who do not exist in the player list.


New Match Performance
* = Required Information
Information
* Match Performance Name
IPL Match 2025
Player of the Match
Match Result
WON
Runs Scored
82
Balls Faced
55
Fours
8
Sixes
3
Cancel
Save & New
Save

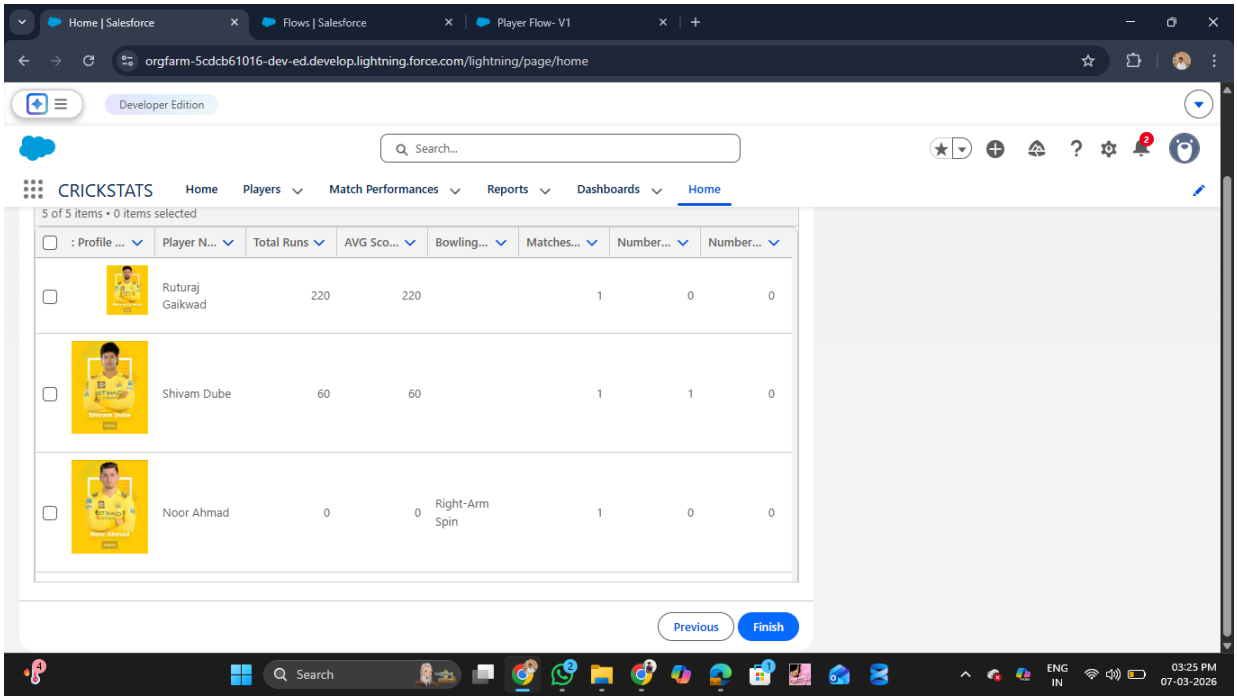

Sixes
3
Overs Bowled
0
Wickets Taken
0
Runs Conceded
0
Maiden Overs
0
Stumpings
0
Catches
1
* Player
Virat Kohli
Cancel
Save & New
Save

- When a user selects or clicks on a **particular match record**, the system shows the **details of the player associated with that match**.





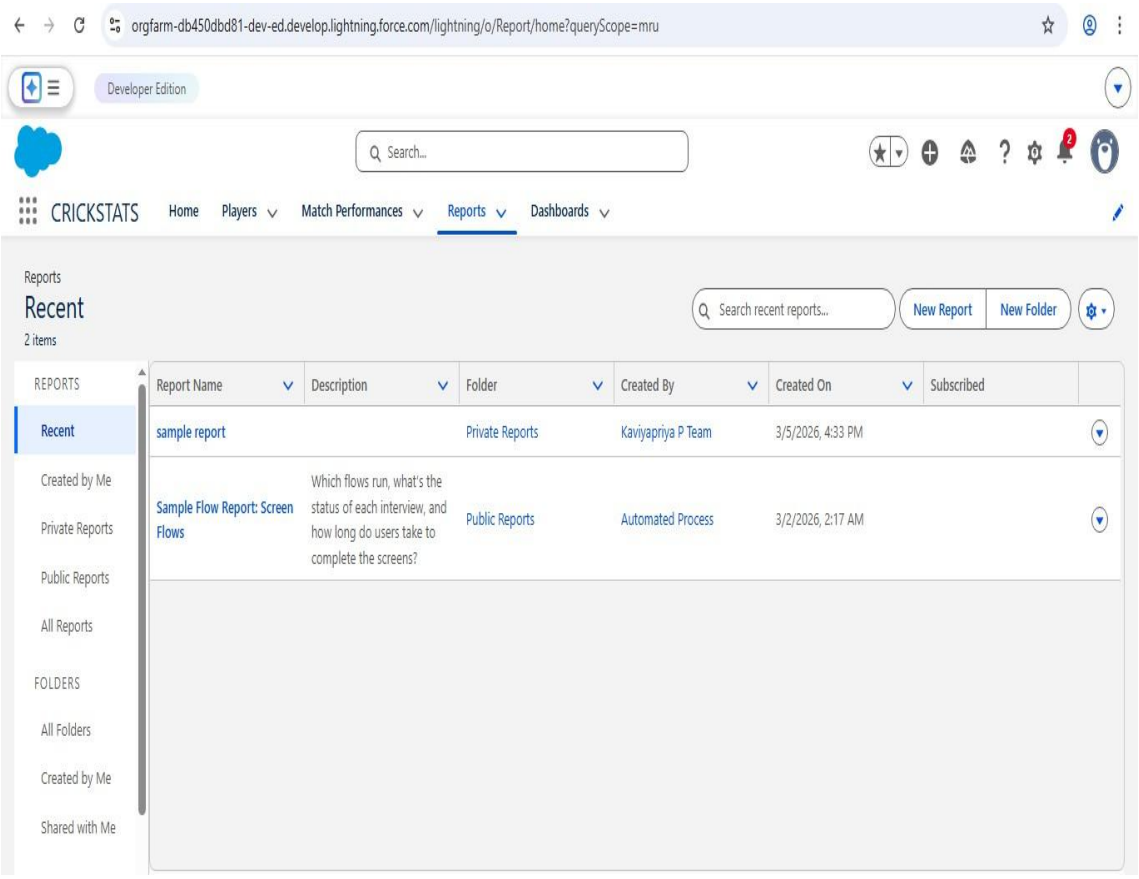
- The page displays important match statistics such as **match result, runs scored, balls faced, fours, sixes, overs bowled, wickets taken, and other performance details**.
- This section helps users **track and analyse the performance of registered players in different matches**.



Reports

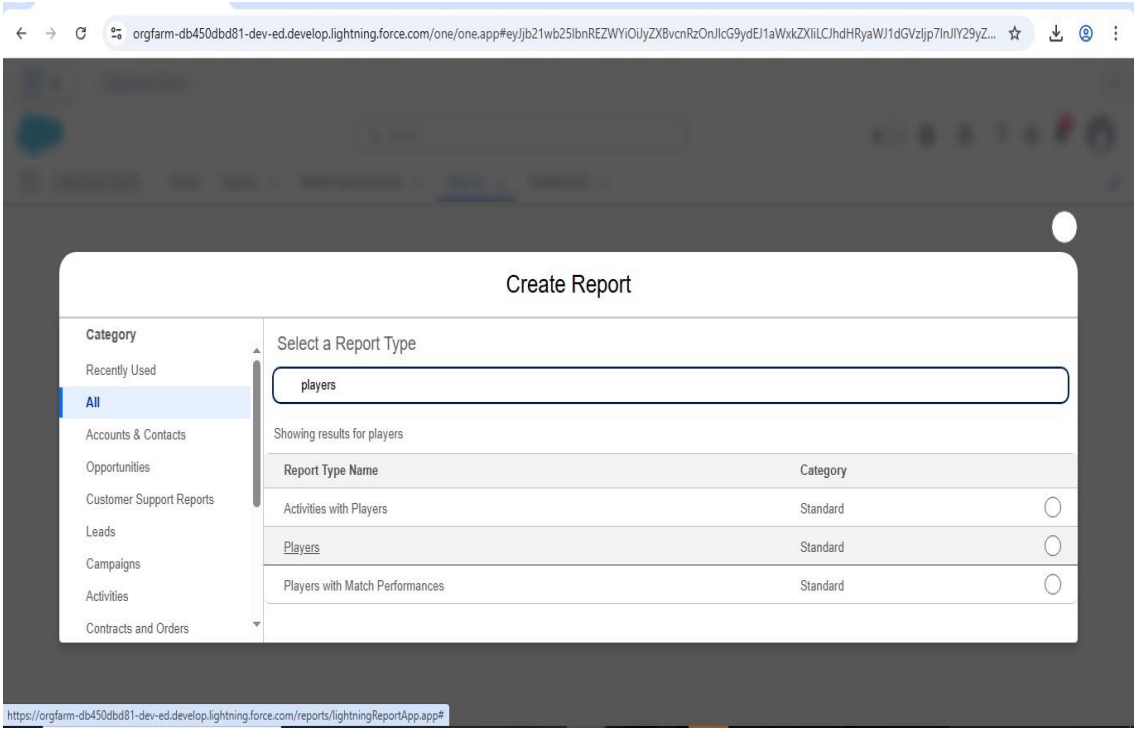
Reports

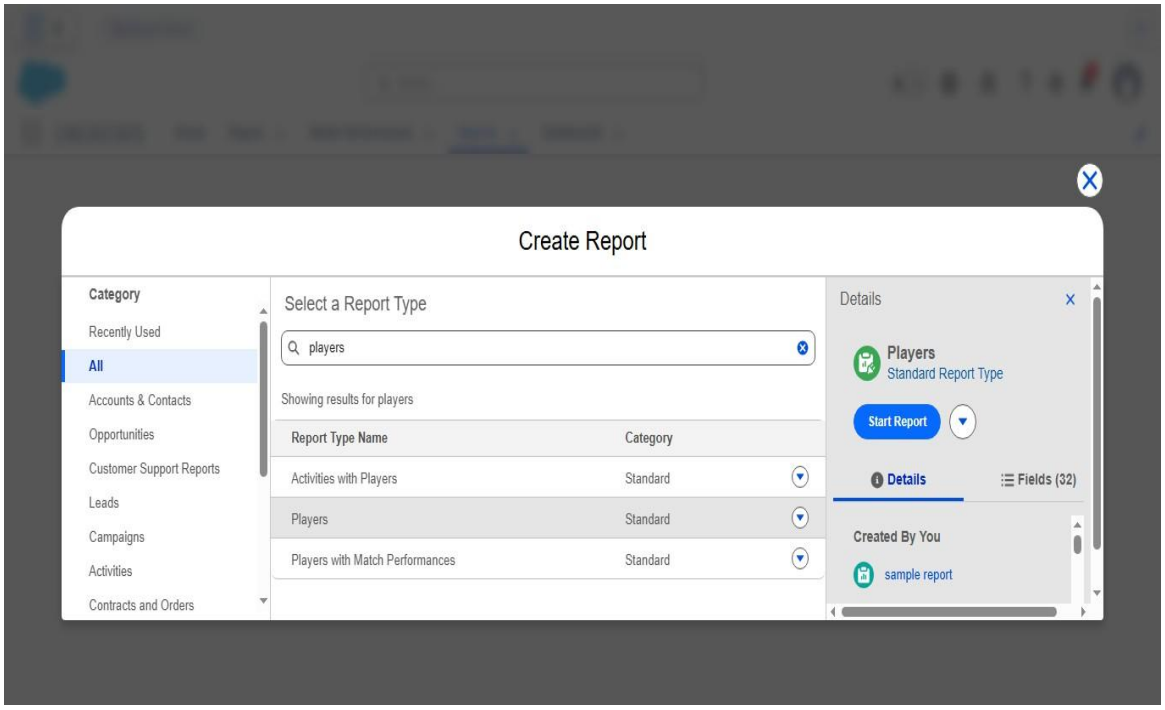
- In the **Reports tab**, the system displays the list of **existing reports** that have already been created in the application.

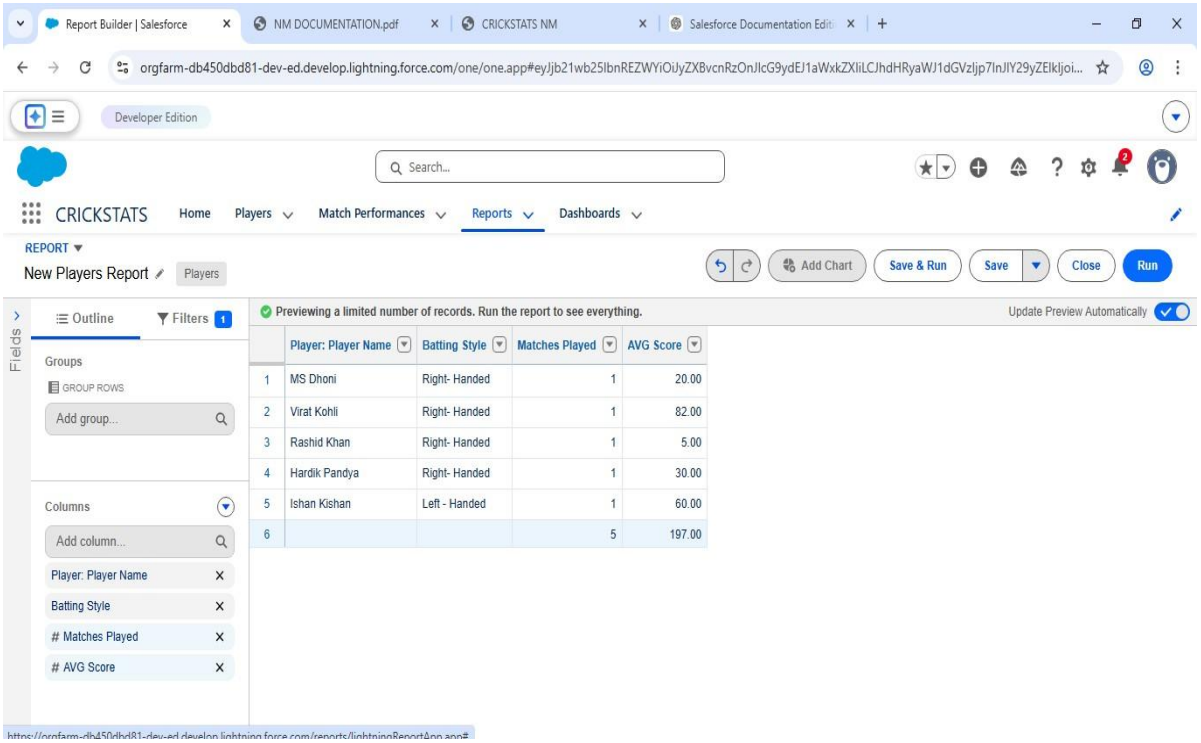


- Users can **view, edit, or delete** these reports based on their requirements.

- The system also allows users to **create new reports** to analyze player and match performance data.






Report Builder | Salesforce
NM DOCUMENTATION.pdf
CRICKSTATS NM
Salesforce Documentation Editi
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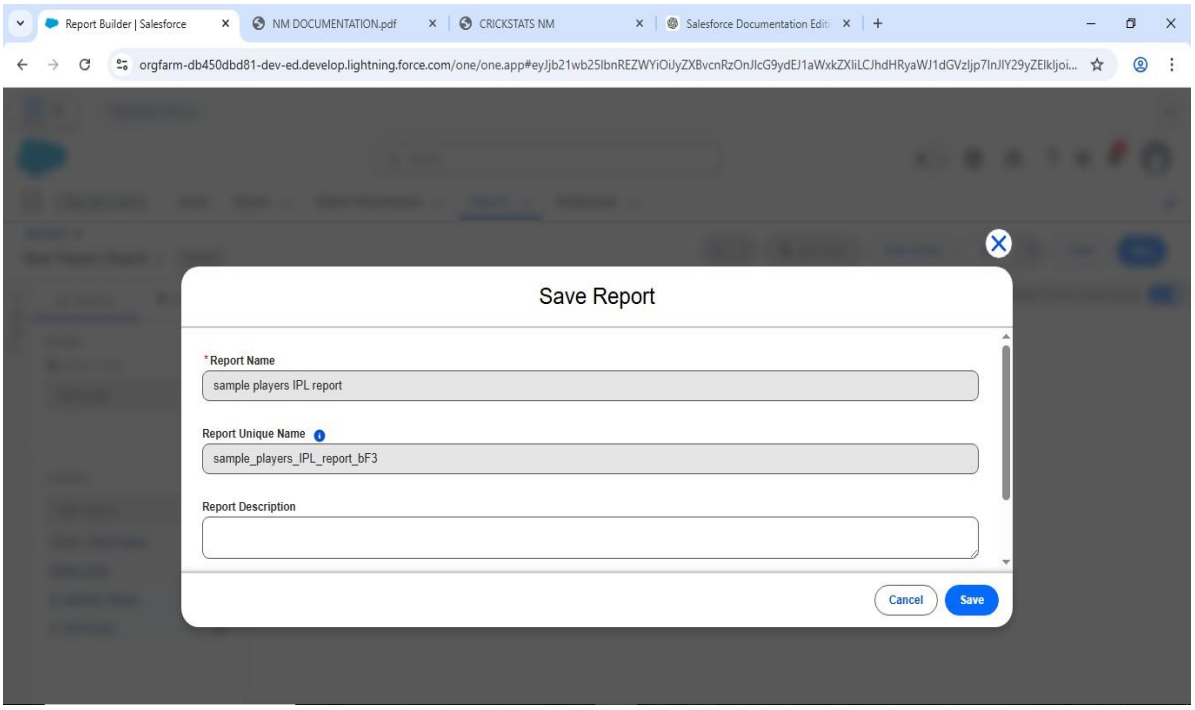
Developer Edition
Search...
CRICKSTATS
Home
Players
Match Performances
Reports
Dashboards
REPORT
New Players Report
Players
Add Chart
Save & Run
Save
Close
Run
Fields
Outline
Filters 1
Groups
GROUP ROWS
Add group...
Columns
Add column...
Player: Player Name
Batting Style
Matches Played
AVG Score
Previewing a limited number of records. Run the report to see everything.
Update Preview Automatically
Player: Player Name
Batting Style
Matches Played
AVG Score
1 MS Dhoni Right- Handed 1 20.00
2 Virat Kohli Right- Handed 1 82.00
3 Rashid Khan Right- Handed 1 5.00
4 Hardik Pandya Right- Handed 1 30.00
5 Ishan Kishan Left - Handed 1 60.00
6 5 197.00


Report Builder | Salesforce
NM DOCUMENTATION.pdf
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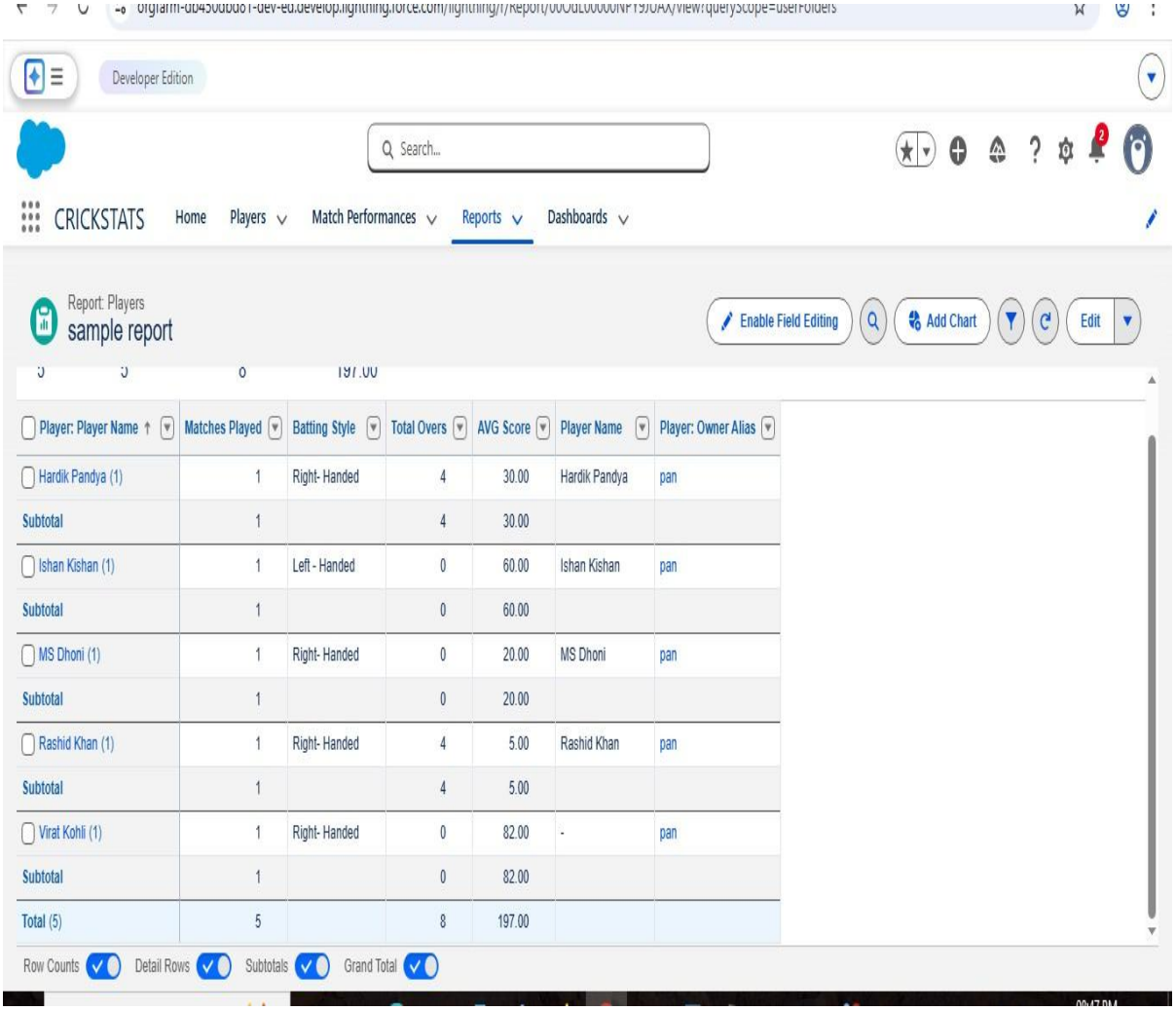
orgfarm-db450dbd81-dev-ed.develop.lightning.force.com/one/one.app#eyJjb21wb25lbnREZWYiOiJyZXBvcnRzOnJlcG9ydEJ1aWxkZXliLCJhdHRyaWJ1dGVzIjp7InJlY29yZEIkIjoi...
Save Report
*Report Name
sample players IPL report
Report Unique Name
sample_players_IPL_report_bF3
Report Description
Cancel
Save

- When a user opens a report, it shows **sample data and report details**, which help in understanding the player statistics.



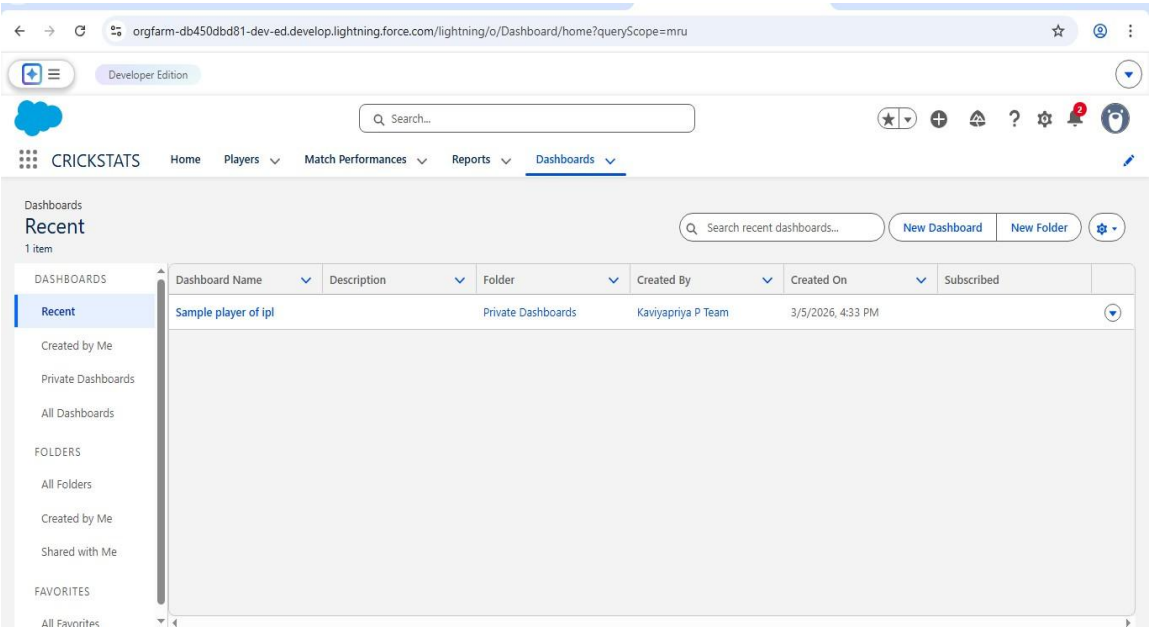
Player: Player Name	Matches Played	Batting Style	Total Overs	AVG Score	Player Name	Player: Owner Alias
Hardik Pandya (1)	1	Right- Handed	4	30.00	Hardik Pandya	pan
Subtotal	1		4	30.00		
Ishan Kishan (1)	1	Left - Handed	0	60.00	Ishan Kishan	pan
Subtotal	1		0	60.00		
MS Dhoni (1)	1	Right- Handed	0	20.00	MS Dhoni	pan
Subtotal	1		0	20.00		
Rashid Khan (1)	1	Right- Handed	4	5.00	Rashid Khan	pan
Subtotal	1		4	5.00		
Virat Kohli (1)	1	Right- Handed	0	82.00	-	pan
Subtotal	1		0	82.00		
Total (5)	5		8	197.00		

- While creating a report, users can **select and filter the fields they want**, arrange how the data should be displayed, and customize the report according to their needs.
- This feature helps users **analyse cricket performance data effectively and generate meaningful insights**.

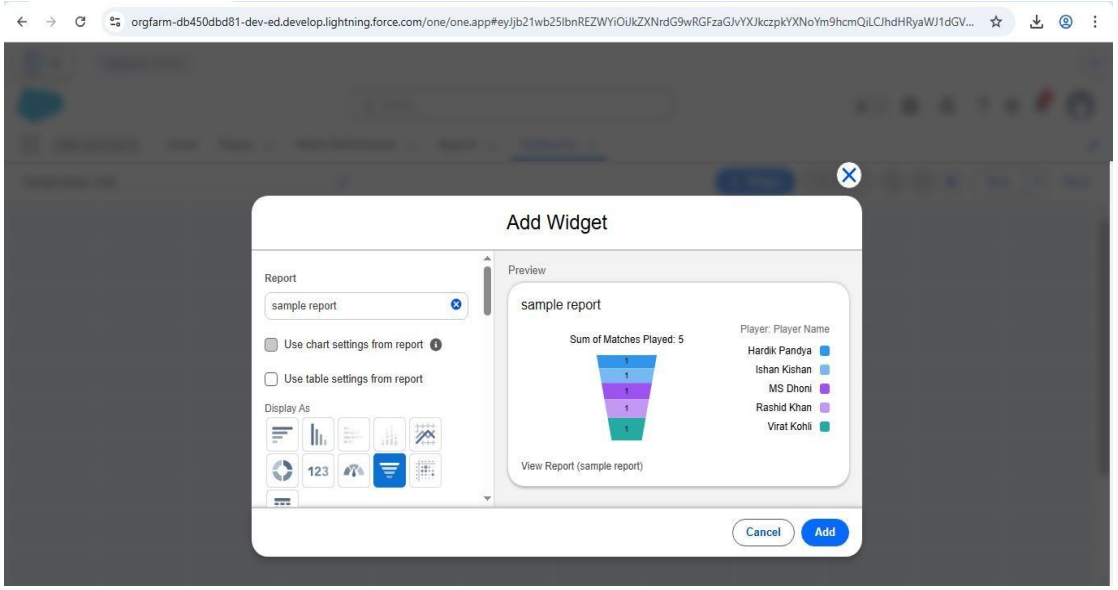
Dashboards

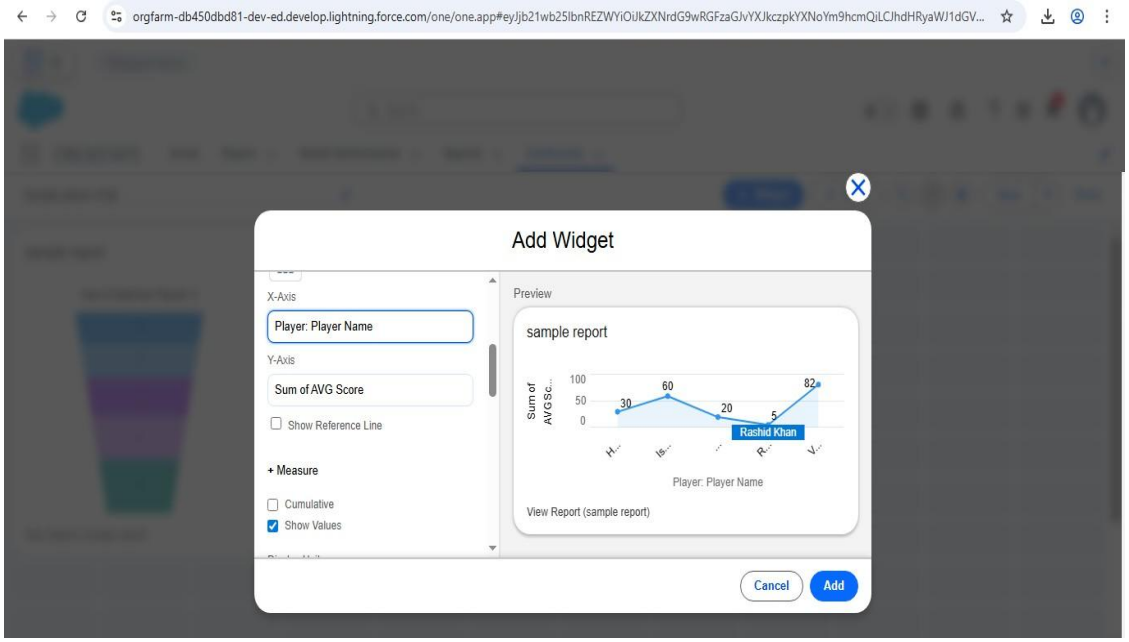
Dashboards

- In the **Dashboard tab**, the system displays player and match statistics in **diagrammatic representations**.

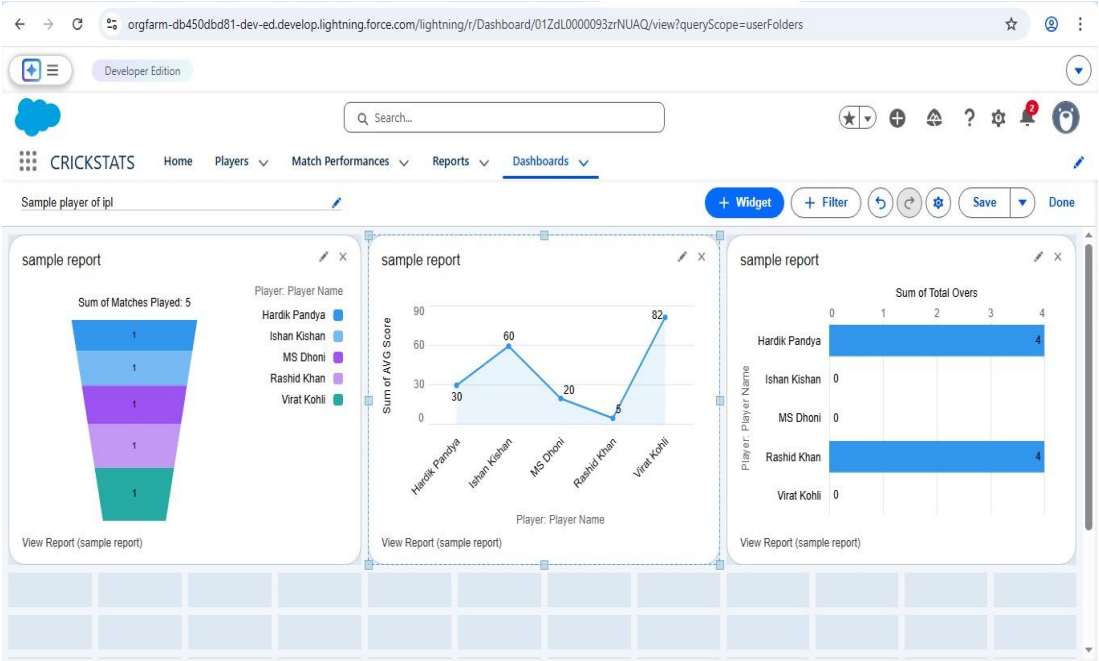


- The data from reports is shown using different visual formats such as **pie charts, bar charts, bar graphs, and metric components**.
- These visualizations make it **easier for users to understand player performance and statistical information**.





- Users can **view, edit, or delete** the existing dashboards based on their requirements.
- The dashboard also allows users to **filter or sort the data** based on different performance metrics such as **average score, number of sixes, runs scored, and other player statistics**.



- This graphical representation helps users **quickly analyze and compare the performance of players**.

Enhancements Implemented

Enhancements Implemented

Several improvements were introduced in the system to enhance its functionality and efficiency.

- Automatic calculation of **batting strike rate**
- Automatic calculation of **bowling economy rate**
- **Performance score ranking** for evaluating player performance
- Use of **roll-up summary fields** to aggregate match statistics
- **Validation rules** to ensure accurate and realistic data entry
- **Record type classification** for organizing different match categories
- Creation of a **custom Lightning App page** for better navigation
- **Screen Flow filtering** to interactively retrieve player data
- Integration of **player profile images**

These enhancements make the **CrickStats application** a more efficient and complete platform for analyzing cricket player performance and statistics.

Conclusion

Conclusion

The **CrickStats** application demonstrates how Salesforce declarative tools can be effectively used to build a complete cricket performance analytics system. The application automates important statistical calculations, maintains data accuracy through validation rules, and presents useful insights using reports and dashboards.

By combining data modeling, automation, and analytical features within the Salesforce Lightning platform, the system provides a scalable and user-friendly solution for managing and analyzing cricket player performance.